Liquidity, Activism Disclosure, and Takeover Premia:

A Theory of Exit, Voice, and Corporate Control

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Abstract

How does secondary-market liquidity shape corporate control when an informed blockholder can choose between exit, quiet engagement, and public activism? I develop a takeover model in which a control-oriented blockholder observes a private signal about the scope for value-creating intervention and chooses among Exit, Hold, Quiet Voice, and threshold-crossing Public Voice. A market maker prices from order flow and disclosure, and a potential bidder conditions entry on the public market signal generated by trading. The model delivers a clean decomposition of post-disclosure prices into fundamentals, standalone intervention value, and takeover premia. In the baseline calibration, expected minority takeover gains are hump-shaped in liquidity: additional noise trading first facilitates intervention and bid incidence, but eventually weakens the inference channel that sustains activism-related premia. Threshold-based disclosure attenuates this channel by shifting probability mass from inferred activism to directly observed activism.

1 Introduction

When an activist investor obtains private information about the scope for value-creating intervention at a portfolio company, she faces a consequential choice: sell her stake and move on, engage quietly behind the scenes, or accumulate shares aggressively enough to cross a regulatory disclosure threshold. Each path carries different risks and rewards, and each sends different signals to the market. How she resolves this dilemma shapes not only her own returns but whether a takeover bid emerges and what minority shareholders ultimately receive. This tension is the classic exit-versus-voice problem of Hirschman (1970), transplanted into financial markets. On one side, liquidity facilitates governance by letting activists accumulate stakes cheaply and reducing free-riding (Maug, 1998). On the other, liquidity makes selling painless, tempting blockholders to exit rather than bear the costs of engagement (Coffee, 1991; Bhide, 1993). The tension deepens once public-market valuations and trading signals influence takeover behavior (Bond et al., 2012; Edmans, Goldstein, et al., 2012). Consequently, the blockholder's choice between exit and voice reverberates beyond her portfolio into the real economy. No existing framework combines these three forces (endogenous governance choice, order-flow inference, and takeover feedback) in a single model.

To fill this gap, I develop a model in which a control-oriented blockholder observes a private signal about intervention value and chooses among four actions: Exit (sell her stake), Hold (retain without engaging), Quiet Voice (engage below the disclosure threshold), or Public Voice (buy additional shares, engage, and trigger mandatory disclosure). A competitive market maker sets prices from discrete order flow and the disclosure flag, while a potential bidder conditions entry on these public market signals. Engagement raises standalone value and strengthens resistance to acquisition, increasing the premium a bidder must offer. The key modeling choice is that disclosure is stake-triggered: activism is directly observable only when the blockholder crosses a regulatory threshold, creating two distinct information regimes within a single equilibrium.

The model yields two main findings. First, numerical equilibrium analysis delivers a nonmonotone relation between liquidity and expected minority takeover gains in the baseline calibration. At low liquidity, noise trading is scarce, making the blockholder's orders highly informative. The market maker largely infers her action, which deters quiet engagement and suppresses bids. As liquidity rises, noise camouflages the activist's trades, encouraging voice and improving bid incidence. However, at high liquidity, the activism-driven component of takeover gains shrinks sharply, so expected minority gains eventually decline. Second, stake-triggered disclosure attenuates this inference channel. Lowering the disclosure threshold shifts probability mass from inferred to observed activism, compressing the activism premium into the observable regime and making it less sensitive to liquidity. The model therefore predicts that the sensitivity of takeover premia to liquidity is lower in strict-disclosure jurisdictions.

These mechanisms speak directly to active policy debates. The United States Schedule 13D threshold at 5% and the United Kingdom FCA threshold at 3% create meaningfully different information environments. The framework clarifies how such cross-country differences reshape bidder beliefs, takeover outcomes, and the information rents available to minority shareholders.

The model's predictions are motivated by well-documented empirical patterns. Brav et al. (2008) show that hedge fund activism often targets firms that subsequently receive acquisition offers. Greenwood and Schor (2009) find that a substantial fraction of activist returns is attributable to takeover outcomes. The free-rider problem that Grossman and Hart (1980) identified in tender offers makes takeover premia the natural welfare object for dispersed minority shareholders, underscoring the importance of understanding how liquidity shapes these outcomes.

The paper proceeds as follows. Section 2 reviews the literature. Section 3 presents the model, and Section 4 characterizes the equilibrium. Section 5 develops comparative statics with numerical analysis. Section 6 presents testable implications. Sections 7 and 8 discuss preliminary welfare accounting and disclosure extensions. Section 9 concludes.

2 Related Literature

This paper connects three strands of the finance literature (blockholder governance, market microstructure, and corporate takeovers) and shows that their interaction produces equilibrium effects that no single strand generates alone.

2.1 Exit vs. Voice in Corporate Governance

The exit-voice trade-off originates with Hirschman (1970). Early applications treated the two as substitutes, arguing that liquidity weakens governance by allowing blockholders to flee rather than fight (Coffee, 1991; Bhide, 1993). Maug (1998) reversed this logic, demonstrating that liquidity can facilitate voice by easing stake accumulation. Edmans (2009) recast exit itself as a governance mechanism, showing that trading on bad private information punishes underperforming managers through equity-linked compensation. Edmans and Manso (2011) extend this to multiple blockholders, while Edmans, V. W. Fang, et al. (2013) provide empirical evidence that liquidity shifts blockholders toward exit and away from voice.[1]

2.2 Market Microstructure and Price Feedback

Microstructure models explain how prices aggregate private information through trading (Kyle, 1985; Glosten and Milgrom, 1985). I build on the tractable discrete order-flow framework of Edmans, Goldstein, et al. (2015), which permits closed-form Bayesian updating and clear price fixed points. Financial prices and other public market signals also shape real economic decisions through feedback channels. Edmans, Goldstein, et al. (2012) demonstrate that stock prices affect takeover outcomes: low prices signal weak fundamentals and invite bids, while high prices deter them. Bond et al. (2012) survey this broader feedback literature.[2] None of these frameworks, however, allow an informed blockholder to choose between exit and voice before public market signals are formed.

2.3 Shareholder Activism and Takeovers

Empirical evidence firmly links activism to corporate control. Brav et al. (2008) document foundational stylized facts regarding activist target selection and subsequent acquisition offers. Greenwood

[1] A common feature of these governance models is continuous trading. The present paper departs from that tradition by adopting discrete order flow, which keeps the inference problem tractable when integrating both disclosure regimes and takeover feedback from public market signals.

[2] Dow et al. (2017) provide foundations for endogenous information production in settings where prices affect real investment.

and Schor (2009) show that a substantial fraction of activist returns traces to these takeover outcomes. The canonical Grossman and Hart (1980) model establishes the free-rider problem that makes the takeover premium the relevant welfare metric for dispersed minority shareholders.[3]

2.4 Relation to This Paper

The closest antecedent is Edmans, Goldstein, et al. (2015), who develop a discrete order-flow model in which public market information feeds back into takeover decisions. That framework shares this paper's tractable Bayesian structure and price fixed points, but differs in three critical respects. First, the blockholder in Edmans, Goldstein, et al. (2015) has an exogenous information advantage and does not choose between alternative governance strategies; in this paper, the blockholder endogenously selects among Exit, Hold, Quiet Voice, and Public Voice, making the governance mode itself a strategic variable. Second, there is no disclosure regime: because the blockholder in Edmans, Goldstein, et al. (2015) does not acquire a stake, there is no threshold-crossing event that partitions the information environment. Here, stake-triggered disclosure splits equilibrium inference into disclosed and nondisclosed branches with sharply different informational content. Third, the absence of endogenous voice means Edmans, Goldstein, et al. (2015) cannot generate the disclosure-attenuation mechanism that constitutes this paper's central contribution, and it does not speak to the baseline numerical nonmonotonicity documented below.

3 Model

The model has five stages that capture the blockholder's information advantage, the market's two-stage inference problem, and the bidder's response.

3.1 Timeline

$t = 0$: Nature draws a standalone fundamental v. The blockholder observes a private signal s about v, which I interpret more broadly as summarizing the scope for profitable intervention.

$t = 1$: The blockholder chooses an action (q, a) from the feasible set $\{(-1, 0), (0, 0), (0, 1), (+1, 1)\}$, where $q \in \{-1, 0, +1\}$ is the trade and $a \in \{0, 1\}$ is engagement. A noise trader submits $z \in \{-1, 0, +1\}$. The market maker observes *only* aggregate order flow $X = q + z$ and clears the blockholder's trade at an anonymous competitive execution price $P_{\text{trade}}(X)$.

$t = 1.2$: Following execution, stake-triggered disclosure $D = \mathbf{1}\{q = +1\}$ is publicly revealed to the market. The secondary market updates the firm's valuation to the fully informed post-disclosure price $P_{\text{post}}(X, D)$.

$t = 1.5$: A potential bidder arrives with probability $\lambda_B \in (0, 1)$, observes (X, D) directly, and draws a private synergy shock ξ, then decides whether to initiate a takeover attempt.

$t = 2$: Payoffs are realized: either the takeover is consummated at the offered price, or the firm remains standalone with (possibly) improved fundamentals due to engagement.

[3] Back et al. (2018) embed an activist in a continuous-time Kyle model to study liquidity and efficiency, but they do not incorporate the feedback loop between order-flow inference and bidder entry conditional on disclosure.

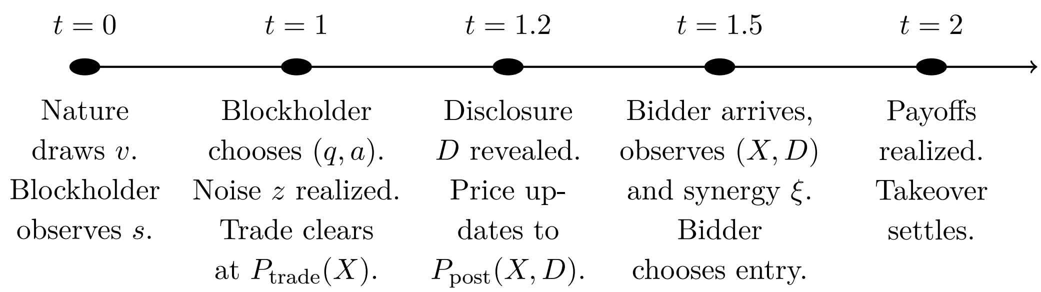

Figure 1: Timeline of the model. The five stages capture the blockholder's information advantage, the market's two-stage inference problem (anonymous execution followed by disclosure), and the bidder's response.

Table 1 in Appendix A provides a quick reference for notation. The remainder of this section introduces each model component in turn, beginning with the blockholder's information environment and ending with the equilibrium concept.

3.2 Fundamentals and Information

The standalone fundamental is $v \sim \mathcal{N}(\mu, \sigma_v^2)$. The blockholder observes a private signal

$$s = v + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2),$$

independent of v. Let $\sigma_s^2 \equiv \sigma_v^2 + \sigma_\varepsilon^2$. The blockholder's posterior mean is linear:

$$\hat{v}(s) \equiv \mathbb{E}[v \mid s] = \mu + \beta(s - \mu), \quad \beta \equiv \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\varepsilon^2} \in (0, 1).$$

Throughout, I interpret higher values of s as combining stronger standalone fundamentals with a stronger intervention thesis: firms with better private signals are not just more valuable on a standalone basis, but also easier for the activist to improve, sell, or reposition credibly. This reduced-form interpretation keeps the Gaussian signal structure while giving the engagement decision an economically meaningful foundation.

The bidder draws an independent synergy shock from a Logistic distribution $\xi \sim \Lambda(0, s_\xi)$. This signal structure gives the blockholder a transient information advantage that she can exploit through trading, engagement, or both.

3.3 Trading and Liquidity

The blockholder has an initial stake of one share (normalization). Her order $q \in \{-1, 0, +1\}$ implies an end-of-$t = 1$ holding of $h = 1 + q$ shares.

Noise trading is discrete:

$$z \in \{-1, 0, 1\}, \quad \mathbb{P}(z = 0) = 1 - \frac{2}{3}\kappa, \quad \mathbb{P}(z = \pm 1) = \frac{\kappa}{3},$$

where $\kappa \in (0, 1)$ indexes noise-trading intensity (market liquidity). Higher κ means more frequent nonzero noise imbalance and therefore weaker inference from order flow.

Total order flow observed by the market maker is

$$X = q + z \in \{-2, -1, 0, 1, 2\}.$$

The blockholder's trading decision is intertwined with her engagement choice, to which I now turn.

3.4 Disclosure Rule

Let $D \in \{0, 1\}$ be a public indicator for whether a stake acquisition crosses a disclosure threshold. I model disclosure as *stake-triggered*: in the discrete framework,

$$D = 1 \iff (q = +1).$$

That is, **if the blockholder acquires an additional share, a threshold-crossing disclosure occurs**. This captures the institutional logic that crossing an ownership threshold triggers a regulatory filing (e.g., Schedule 13D), independent of whether the filing explicitly reflects intent. In the baseline model I treat threshold crossing as a public intervention choice by a control-oriented activist, so the only threshold-crossing action on path is Public Voice $(+1, 1)$. Passive threshold-crossing accumulation is discussed only as an appendix comparison and is not part of the baseline equilibrium object. Below-threshold engagement remains hidden.

Timing of disclosure and anonymous accumulation. The disclosure indicator D is mechanically triggered by the blockholder's action (q, a), but its revelation to the market is delayed until after the trade clears. Specifically, the sequence from $t = 1$ to $t = 1.2$ is:

(i) The blockholder chooses (q, a), locking in her ultimate disclosure status $D = \mathbf{1}\{q = +1\}$.

(ii) Noise trader submits z; total order flow $X = q + z$ is realized.

(iii) The market maker observes *only* X and clears the blockholder's trade at the anonymous execution price $P_{\text{trade}}(X)$.

(iv) Following execution, the disclosure indicator D is publicly filed, and the secondary market updates to the fully informed post-disclosure price $P_{\text{post}}(X, D)$.

This timing reflects standard institutional mechanics (e.g., the SEC Rule 13d-1 filing window): activists accumulate shares anonymously in the open market limit order book before the regulatory filing reveals their presence and intent.

3.5 Engagement Technology (Voice)

If the blockholder chooses engagement $a = 1$, she pays a private, signal-dependent cost $C(s) > 0$ at $t = 1$. This cost is observed by the blockholder when she observes s but not by the market maker or bidder. I assume engagement costs are decreasing in the signal: a blockholder with a stronger intervention thesis finds it easier to build a compelling case for change. I parameterize the cost as

$$C(s) = C_0 \exp\left(-\chi \frac{s-\mu}{\sigma_s}\right),$$

where $C_0 > 0$ is the cost at the prior mean and $\chi \geq 0$ controls cost sensitivity to signal quality (Assumption (A2)). The assumption that engagement costs are decreasing in the signal reflects the economics of persuasion and sale initiation. A blockholder with more favorable private information can point to clearer restructuring opportunities, favorable peer comparisons, or more credible evidence that a strategic review would unlock value, substantially lowering the friction of persuading the board or rallying other institutions. The exponential form is chosen for analytical convenience to ensure non-negativity, but the qualitative results require only strict positivity and a monotone decrease in s. This monotonicity assumption is economically substantive. If costs were instead increasing in s, the cutoff ordering could invert, pushing the blockholder toward exit on good news and voice on bad news. The decreasing cost structure captures the baseline paradigm in which a stronger intervention thesis makes activism easier to execute.

This monotone, strictly positive specification ensures that neither engagement nor passivity is always optimal when the blockholder holds her initial stake, yielding an interior cutoff structure (Assumption (A1)).[4]

Engagement succeeds with probability $\rho \in (0, 1]$ and affects outcomes in two ways when successful:

Outside option improvement. If engagement succeeds and no takeover is consummated, the standalone payoff is increased by $\Delta > 0$ at $t = 2$. If engagement fails, standalone payoff remains v.

Bargaining/resistance premium. If engagement succeeds and a takeover is consummated, the required premium increases from m_0 to m_1, where $m_1 > m_0 \geq 0$. The wedge captures bargaining power, defensive tactics, or any friction that raises the price needed to complete the acquisition when an engaged blockholder is present. Failed engagement yields the baseline premium m_0. I interpret m_0 and m_1 as *per-share takeover premia* above the expected standalone fundamental value (so the consummated offer satisfies $b = \hat{V}(X, D) + m^R(a)$); they are distinct from $\hat{v}(s)$, the posterior mean of fundamentals.

Risk-neutral simplification. Since success is unobserved before $t = 2$ and all parties are risk-neutral, I define expected engagement effects:

$$\tilde{\Delta} \equiv \rho\Delta, \qquad \tilde{m} \equiv m_0 + \rho(m_1 - m_0).$$

Then engagement ($a = 1$) yields expected standalone improvement $\tilde{\Delta}$ and expected premium wedge $\tilde{m}$. I assume the premium wedge satisfies $\tilde{m} > m_0$, which requires $\rho(m_1 - m_0) > 0$ (Assump-

[4] Sufficient conditions for interior cutoffs include $C(\mu) < \delta(\tilde{\Delta} + (\tilde{m} - m_0))$ (engagement is worthwhile at the prior mean) and $\lim_{s\to-\infty} C(s) > \delta(\tilde{\Delta} + (\tilde{m} - m_0))$ (engagement is dominated for sufficiently low signals), together with the monotonicity of $C(s)$. I also assume that stake building (Public Voice) is not always dominated when engagement occurs, so that disclosure is triggered for sufficiently high signals. When $\chi > 0$, cost heterogeneity supports an interior Hold region; when $\chi = 0$, costs are constant and the equilibrium collapses to the two-cutoff case.

tion (A3)). This ensures that engagement has real consequences for takeover outcomes; an engaged blockholder can extract higher premia through an improved bargaining position or defensive tactics.

3.6 Bidder Entry and Bidding Terms

The bidder follows an entry rule conditional on the market's inference. A potential bidder arrives with exogenous Poisson probability $\lambda_B \in (0,1)$ at $t = 1.5$. The bidder observes the same public trading and disclosure history as the market, summarized by the sufficient statistic (X, D), together with a private synergy shock ξ. This reduced-form information structure avoids any reliance on price injectivity. In the baseline action set, the disclosed branch $D = 1$ corresponds to Public Voice and hence to $a = 1$, so X reveals only the noise realization and carries no information about fundamentals beyond that public intervention signal (see Appendix B.4).

I assume that synergies and costs satisfy $\bar{S} - K > m_0 - \mu$ and $\bar{S} - K < m_1 + (\mu + 3\sigma_v)$ (Assumption (A4)). These bounds ensure that takeover bids occur with interior probability. Crucially, successful engagement ($a = 1$) strips away managerial entrenchment, forcing a competitive auction. This raises the expected fundamental synergy to $\bar{S} + \Delta_S$. Thus, activism *fundamentally facilitates* the sale.

The bidder anchors their takeover offer to the target's expected standalone fundamental value, $\hat{V}(X, D) \equiv \mathbb{E}[v \mid X, D] + \tilde{\Delta}\pi(X, D)$, plus the *inferred* premium wedge. Because the bidder does not perfectly observe a when $D = 0$, the payout is tied to the market's Bayesian expectation. The expected per-share offer required to consummate the acquisition is:

$$b(X, D) = \hat{V}(X, D) + \bar{m}(X, D),$$

where $\bar{m}(X, D) = m_0 + (\tilde{m} - m_0)\pi(X, D)$ is the expected premium wedge, and $\pi(X, D) = \mathbb{P}(a = 1 \mid X, D)$. By anchoring the offer to the fundamental standalone value rather than the anticipatory market prices $P_{\text{trade}}(X)$ or $P_{\text{post}}(X, D)$, the model mathematically prevents a pathological infinite recursion of pricing a premium-on-a-premium.

Conditional on arrival, the bidder's expected net deal surplus is:

$$\Pi_B(X, D) = (\bar{S} + \pi(X, D)\Delta_S) + \xi - b(X, D) - K,$$

where $K > 0$ is a fixed bidding cost. The conditional probability that an arriving bidder initiates a consummated takeover is:

$$\tilde{p}(X, D) \equiv 1 - \Lambda\left(\frac{\hat{V}(X, D) + \bar{m}(X, D) + K - (\bar{S} + \pi(X, D)\Delta_S)}{s_\xi}\right), \tag{1}$$

where $\Lambda(\cdot)$ is the standard logistic cumulative distribution function, reflecting a logistic synergy shock $\xi \sim \Lambda(0, s_\xi)$ (introduced below). The unconditional bid probability is $p(X, D) = \lambda_B \cdot \tilde{p}(X, D)$. I additionally assume a bounded bidder arrival rate $\lambda_B \leq 1/2$ (Assumption (A7)), which is trivially satisfied in U.S. M&A data where annual bid incidences are single-digit percentages, and ensures the single-crossing condition used in Proposition 1.

To ensure that the efficient pricing of the fundamental value run-up makes acquisitions more expensive on net despite synergy gains, I assume $\tilde{\Delta} + \tilde{m} - m_0 > \Delta_S$ (Assumption (A5)).

If the takeover is consummated, the *realized* required premium is a function of the realized action, $m^R(a) \equiv m_0 + a \cdot (\tilde{m} - m_0)$, yielding a per-share offer of:

$$b(X, D, a) = \hat{V}(X, D) + m^R(a).$$

3.7 Terminal Payoff and Price Formation

Let $\delta = \exp(-r\tau)$ (approximately $1/(1+r)$ for short horizons) be the discount factor between the pricing/entry date and payoff/settlement at $t = 2$, where τ is the time between these dates (Cochrane, 2005; Duffie, 2010).[5]

If no takeover is consummated, per-share terminal payoff is $y^{\text{stand}} = v + a\tilde{\Delta}$. If a takeover is consummated, per-share payoff is $y^{\text{M\&A}} = b(X, D, a) = \hat{V}(X, D) + m^R(a)$.

Therefore, the per-share terminal payoff is

$$Y = \mathbf{1}\{\text{bid}\} \cdot (\hat{V}(X, D) + m^R(a)) + (1 - \mathbf{1}\{\text{bid}\}) \cdot (v + a\tilde{\Delta}),$$

where $\mathbf{1}\{\text{bid}\}$ denotes a *consummated* takeover rather than a mere offer. Bargaining, board resistance, and regulatory frictions are captured in reduced form by the premium wedge $m^R(\cdot)$ and the fixed cost K.

Conditional on (X, D), the expected takeover payout equals $\hat{V}(X, D) + \bar{m}(X, D)$.

The market maker is competitive. Because trading occurs prior to the regulatory filing, the anonymous execution price paid by the blockholder at $t = 1$ is the expectation over latent disclosure states conditional purely on aggregate order flow:

$$P_{\text{trade}}(X) = \delta\, \mathbb{E}[Y \mid X] = \sum_{d \in \{0,1\}} \mathbb{P}(D = d \mid X) P_{\text{post}}(X, d). \tag{2}$$

Following the regulatory filing at $t = 1.2$, the secondary market perfectly observes D and updates to the post-disclosure expected terminal payoff:

$$P_{\text{post}}(X, D) = \delta\, \mathbb{E}[Y \mid X, D]. \tag{3}$$

Because the bidder's entry condition (1) is cleanly anchored to the expected standalone fundamental value $\hat{V}(X, D)$ rather than the anticipatory market prices, the bid probability is strictly independent of both $P_{\text{trade}}(X)$ and $P_{\text{post}}(X, D)$. This resolves the recursive premium-on-a-premium problem and leaves the pricing equations feed-forward once beliefs are specified.

3.8 Blockholder Payoff and Equilibrium Concept

Conditional on signal s, the blockholder chooses (q, a) to maximize expected present value. If she submits q and ends with $h = 1 + q$ shares, her $t = 1$ expected utility is:

$$U(q, a \mid s) = \mathbb{E}\Big[-q \cdot P_{\text{trade}}(X) + \delta \cdot h \cdot Y - a \cdot C(s) \,\Big|\, s, q, a \Big],$$

[5] If one measures payoffs in time-1 present-value units, δ can be set to one and later reintroduced by scaling. I keep δ explicit throughout.

where $-qP_{\text{trade}}(X)$ is the net cash flow from anonymous trading at $t = 1$ (selling $q = -1$ yields $+P_{\text{trade}}$; buying $q = +1$ yields $-P_{\text{trade}}$).

Definition 1 (Perfect Bayesian Equilibrium)**.** *A* Perfect Bayesian Equilibrium *consists of:*

(i) a blockholder strategy mapping s into (q, a),

(ii) post-disclosure beliefs $\pi(X, D) = \mathbb{P}(a = 1 \mid X, D)$ and pre-disclosure beliefs $\mathbb{P}(D = d \mid X)$ consistent with Bayes' rule,

(iii) a bidder entry strategy satisfying (1)*,*

(iv) a post-disclosure competitive price schedule $P_{\text{post}}(X, D)$ satisfying (3)*, and*

(v) an anonymous execution price schedule $P_{\text{trade}}(X)$ satisfying (2)*.*

For interior liquidity values $\kappa \in (0, 1)$, noise trading z has full support on $\{-1, 0, +1\}$, so every aggregate order flow $X \in \{-2, -1, 0, 1, 2\}$ occurs with positive probability. The threshold-crossing disclosure $D = 1$ occurs strictly if and only if $q = +1$, mapping uniquely to $X \in \{0, 1, 2\}$. Consequently, Bayes' rule pins down both $\pi(X, D)$ and the latent disclosure probabilities $\mathbb{P}(D = d \mid X)$ on all reached information sets; off-path beliefs are only required to complete the equilibrium specification.[6]

4 Equilibrium Characterization

4.1 Threshold Structure with Three Cutoffs

Because the paper studies a control-oriented activist, the baseline action set contains only one threshold-crossing action, Public Voice. The next lemma is therefore a supporting result rather than a primitive of the equilibrium definition: it shows that passive threshold crossing is dominated for sufficiently high signals and is not used in the baseline calibration.

Lemma 1 (Supporting Note on Passive Threshold Crossing)**.** *Fix an arbitrary market belief system $\pi(X, 1) \in [0, 1]$ following a regulatory filing $D = 1$. There exists a finite threshold $\bar{s} \in \mathbb{R}$ such that, for all $s \geq \bar{s}$, Public Voice strictly dominates Quiet Accumulation:*

$$U(+1, 1 \mid s) \;>\; U(+1, 0 \mid s).$$

In the baseline calibration and in the parameter sweeps reported in Section 5, the Public Voice cutoff satisfies $k_D \geq \bar{s}$. This supports the baseline interpretation that a reached disclosed state represents public intervention rather than passive threshold crossing.

Proof. See Appendix B.1.

Under that baseline interpretation, the blockholder's optimal active strategy has a natural ordering: she exits on bad news, holds passively on mildly negative news, engages quietly on moderate news, and goes public on good news. The next result makes this precise.

[6] Throughout, I maintain Assumptions (A1)–(A7) (the Standing Assumptions). These ensure interior action regions and nontrivial takeover outcomes. Proposition 4 gives an existence result under explicit compactness and continuity conditions on the cutoff mapping, while Assumption (A6) is treated as a numerical regularity condition rather than an analytic uniqueness theorem.

Proposition 1 (Monotone Cutoff Characterization)**.** *In any monotone-cutoff Perfect Bayesian Equilibrium, the blockholder follows cutoff rules with (weakly) ordered thresholds* $k_1 \le k_0 \le k_D$*:*

$$(q,a) = \begin{cases} (-1,0) & s < k_1 \quad \textit{(Exit)}, \\ (0,0) & k_1 \le s < k_0 \quad \textit{(Hold)}, \\ (0,1) & k_0 \le s < k_D \quad \textit{(Quiet Voice)}, \\ (+1,1) & s \ge k_D \quad \textit{(Public Voice)}. \end{cases}$$

The corresponding disclosure outcomes are: $D = 1$ *if and only if* $q = +1$ *(Public Voice), and* $D = 0$ *otherwise.*

Proof. See Appendix B.2.

Remark 1 (Collapse of the Hold Region). The existence of an interior Hold region ($k_1 < s < k_0$) requires the baseline engagement cost C_0 to be sufficiently large. The blockholder prefers passive retention over quiet engagement at marginal signals if and only if $C(s) > \delta \mathbb{E}_z[p(X,0)(\tilde{m} - m_0) + (1 - p(X,0))\tilde{\Delta}]$. Because the unconditional bid probability p is empirically small, the right-hand side is well approximated by $\delta\tilde{\Delta}$. If $C_0 \le \delta\tilde{\Delta}$, the expected return to active engagement strictly dominates passivity for all active signals, causing the Hold region to structurally collapse ($k_0 = k_1$). This is a parametric reality of the blockholder's cost function, not a structural flaw; as demonstrated in Section 5, higher baseline frictions cleanly restore the Hold region without altering the underlying order-flow inference mechanism or the baseline numerical liquidity patterns.

The ordering reflects how the blockholder's incentives shift with her private signal. For very low signals, she holds unfavorable information about fundamentals and prefers to sell, monetizing her negative information before it reaches prices.

For somewhat better signals, fundamentals are not bad enough to warrant selling at a discount, but engagement is not yet worthwhile given its cost. The blockholder holds passively.

For intermediate signals, engagement improves expected value enough to justify the campaign cost $C(s)$, but stake building remains unattractive: buying additional shares costs more than the benefit of higher ownership. The blockholder engages quietly.

For the highest signals, buying additional shares allows the blockholder to internalize more of the engagement gains, and the discrete stake increase triggers public disclosure. This is the Public Voice region.

When engagement costs are constant (i.e., $\chi = 0$), the Hold region may collapse, yielding a simpler two-cutoff structure with $k_0 = k_1$. The analysis accommodates this case by allowing weak inequalities among cutoffs.

4.2 Action Probabilities

Given cutoffs (k_1, k_0, k_D), define standardized cutoffs $\alpha_i \equiv (k_i - \mu)/\sigma_s$ for $i \in \{1, 0, D\}$. The unconditional action probabilities are:

$$\begin{aligned}
\omega_E &\equiv \mathbb{P}(s < k_1) = \Phi(\alpha_1), \\
\omega_H &\equiv \mathbb{P}(k_1 \le s < k_0) = \Phi(\alpha_0) - \Phi(\alpha_1), \\
\omega_Q &\equiv \mathbb{P}(k_0 \le s < k_D) = \Phi(\alpha_D) - \Phi(\alpha_0), \\
\omega_P &\equiv \mathbb{P}(s \ge k_D) = 1 - \Phi(\alpha_D),
\end{aligned}$$

corresponding to Exit, Hold, Quiet Voice, and Public Voice.

4.3 Conditional Means of Fundamentals

Let $\lambda_L(\alpha) \equiv \phi(\alpha)/\Phi(\alpha)$ and $\lambda_U(\alpha) \equiv \phi(\alpha)/(1 - \Phi(\alpha))$ denote inverse Mills ratios. The conditional means of v in each signal region are:

$$\begin{aligned}
\mu_E &\equiv \mathbb{E}[v \mid s < k_1] = \mu - \beta\sigma_s\lambda_L(\alpha_1), \\
\mu_H &\equiv \mathbb{E}[v \mid k_1 \le s < k_0] = \mu + \beta\sigma_s\frac{\phi(\alpha_1) - \phi(\alpha_0)}{\Phi(\alpha_0) - \Phi(\alpha_1)}, \\
\mu_Q &\equiv \mathbb{E}[v \mid k_0 \le s < k_D] = \mu + \beta\sigma_s\frac{\phi(\alpha_0) - \phi(\alpha_D)}{\Phi(\alpha_D) - \Phi(\alpha_0)}, \\
\mu_P &\equiv \mathbb{E}[v \mid s \ge k_D] = \mu + \beta\sigma_s\lambda_U(\alpha_D).
\end{aligned}$$

4.4 Bayesian Posteriors

Disclosure creates a sharp asymmetry in what the market can learn: when activism is disclosed, inference is trivial; when it is not, the market must extract engagement probabilities from noisy order flow. The next result characterizes the posterior beliefs that sustain equilibrium pricing.

Proposition 2 (Posterior Engagement Probabilities)**.** *For any information set (X, D) that is reached with positive probability in equilibrium, the posterior probability of engagement $\pi(X, D) = \mathbb{P}(a = 1 \mid X, D)$ satisfies:*

(a) ***Disclosed states:*** *If $D = 1$, then $\pi(X, 1) = 1$ for all compatible $X \in \{0, 1, 2\}$.*

(b) ***Nondisclosed states:*** *If $D = 0$, let $p_0 \equiv 1 - \frac{2}{3}\kappa$ and $p_1 \equiv \frac{\kappa}{3}$. Then:*

$$\begin{aligned}
\pi(1, 0) &= \frac{\omega_Q}{\omega_H + \omega_Q}, \\
\pi(-1, 0) &= \frac{\omega_Q\, p_1}{(\omega_H + \omega_Q)\, p_1 + \omega_E\, p_0}, \\
\pi(0, 0) &= \frac{\omega_Q\, p_0}{(\omega_H + \omega_Q)\, p_0 + \omega_E\, p_1}, \\
\pi(-2, 0) &= 0.
\end{aligned}$$

Proof. See Appendix B.3.

When $D = 1$, the inference problem collapses entirely: under the baseline action set, disclosure reveals that the blockholder chose Public Voice, which implies engagement with certainty. The market knows public intervention has occurred, though the ultimate success of the campaign remains uncertain until $t = 2$.

When $D = 0$, the market faces a genuine inference problem. It observes only aggregate order flow X and must form beliefs about whether the blockholder is engaged (Quiet Voice) or not (Exit or Hold). The key insight is that Hold and Quiet Voice both involve $q = 0$, so they generate identical order-flow distributions; the market cannot distinguish between them from order flow alone. Instead, the posterior split between these actions is pinned down by their prior probabilities (ω_H, ω_Q), which depend on the equilibrium cutoffs.

This is where liquidity enters the inference channel: higher κ changes the noise distribution and thereby affects how order flow X is interpreted. The posteriors $\pi(X, 0)$ inherit this liquidity dependence, transmitting it to prices and takeover premia.

4.5 Conditional Expectation of Fundamentals Given (X, D)

For pricing, we need $\mathbb{E}[v \mid X, D]$. Since v is independent of noise z, only the action regions compatible with (X, D) matter.

For $D = 1$: Only Public Voice is possible:

$$\mathbb{E}[v \mid X, D = 1] = \mu_P \quad \text{for } X \in \{0, 1, 2\}.$$

For $D = 0$: Mix over Exit, Hold, and Quiet Voice with Bayesian weights. Let $p_0 \equiv 1 - \frac{2}{3}\kappa$ and $p_1 \equiv \frac{\kappa}{3}$. Then:

$$\begin{aligned}
\mathbb{E}[v \mid -2, 0] &= \mu_E, \\
\mathbb{E}[v \mid 1, 0] &= \frac{\omega_H \mu_H + \omega_Q \mu_Q}{\omega_H + \omega_Q}, \\
\mathbb{E}[v \mid -1, 0] &= \frac{(\omega_H \mu_H + \omega_Q \mu_Q)\, p_1 + \omega_E \mu_E\, p_0}{(\omega_H + \omega_Q)\, p_1 + \omega_E\, p_0}, \\
\mathbb{E}[v \mid 0, 0] &= \frac{(\omega_H \mu_H + \omega_Q \mu_Q)\, p_0 + \omega_E \mu_E\, p_1}{(\omega_H + \omega_Q)\, p_0 + \omega_E\, p_1}.
\end{aligned}$$

4.6 Equilibrium Prices

The competitive pricing conditions evaluate directly as pure feed-forward expectations. Because the bid probability $p(X, D)$ relies on the fundamental standalone value rather than recursively on market prices, the post-disclosure equilibrium valuation is strictly determined by the market's Bayesian inference. For each post-disclosure information set (X, D) and posterior $\pi(X, D)$, the unique secondary market price explicitly satisfies:

$$P_{\text{post}}(X, D) = \delta\Big((1 - p(X, D)) \cdot \hat{V}(X, D) + p(X, D) \cdot (\hat{V}(X, D) + \bar{m}(X, D))\Big), \tag{4}$$

where $\hat{V}(X,D) \equiv \mathbb{E}[v \mid X,D] + \tilde{\Delta} \cdot \pi(X,D)$ is the expected standalone value, and $p(X,D) = \lambda_B \cdot \tilde{p}(X,D)$. This simplifies algebraically to:

$$P_{\text{post}}(X,D) = \delta\Big(\hat{V}(X,D) + p(X,D) \cdot \bar{m}(X,D)\Big). \tag{5}$$

The post-disclosure price is exactly the expected standalone fundamental value plus the expected probability-weighted takeover premium. When a takeover is unlikely ($p(X,D) \approx 0$), this reduces to $P_{\text{post}}(X,D) \approx \delta\,\hat{V}(X,D)$.

The anonymous execution price paid by the blockholder at $t = 1$ is the Bayesian expectation over these post-disclosure states, shielding her accumulation from immediate public front-running:

$$P_{\text{trade}}(X) = \sum_{d \in \{0,1\}} \mathbb{P}(D = d \mid X)\, P_{\text{post}}(X,d). \tag{6}$$

The latent disclosure probability in (6) follows from Bayes' rule:

$$\mathbb{P}(D = 1 \mid X) = \frac{\omega_P\, \mathbb{P}(z = X-1)}{\omega_E\, \mathbb{P}(z = X+1) + (\omega_H + \omega_Q)\, \mathbb{P}(z = X) + \omega_P\, \mathbb{P}(z = X-1)}.$$

With bounded support, $\mathbb{P}(D = 1 \mid X = 2) = 1$, $\mathbb{P}(D = 1 \mid X \in \{-2,-1\}) = 0$, and only $X \in \{0,1\}$ mix across disclosure states.

4.7 Price Decomposition and Activism Premium

Building on the feed-forward pricing equation (5), I decompose the post-disclosure secondary market price into interpretable components.

Proposition 3 (Post-Disclosure Price Decomposition)**.** *The unique post-disclosure equilibrium price $P_{\text{post}}(X,D)$ satisfies*

$$P_{\text{post}}(X,D) = \delta\Big(\mathbb{E}[v \mid X,D] + \tilde{\Delta} \cdot \pi(X,D) + p(X,D) \cdot \bar{m}(X,D)\Big),$$

where $p(X,D)$ is the unconditional bid probability, $\pi(X,D) = \mathbb{P}(a = 1 \mid X,D)$, and $\bar{m}(X,D) = m_0 + (\tilde{m} - m_0)\pi(X,D)$.

The terms involving $\pi(X,D)$ constitute a structural "activism premium":

- ***Standalone channel:*** *$\tilde{\Delta}\pi(X,D)$ reflects the fully capitalized unconditional expected value improvement from engagement.*
- ***Takeover channel:*** *$p(X,D) \cdot (\tilde{m} - m_0)\pi(X,D)$ reflects the expected* incremental *takeover premium attributable to the activist's bargaining friction (over and above the baseline m_0).*

By the law of iterated expectations, the anonymous execution price $P_{\text{trade}}(X)$ inherits this exact decomposition, weighted by the market's pre-disclosure inference regarding the latent regulatory state D.

Proof. See Appendix B.6.

Activism affects post-disclosure prices through two distinct channels. The *standalone channel* operates when no takeover occurs: an engaged blockholder improves firm value by $\tilde{\Delta}$ in expectation, and this improvement is capitalized into the share price. The *takeover channel* operates when a bid arrives: the bidder must pay a higher premium to acquire a firm with an engaged blockholder, and this expected higher premium is also reflected in the pre-bid secondary market price.

The asymmetry between disclosed and nondisclosed states is central to the mechanism. In disclosed states ($D = 1$), engagement is known with certainty, so $\pi(X, 1) = 1$ and the activism premium is fully priced into P_{post}. In nondisclosed states ($D = 0$), the posterior $\pi(X, 0)$ depends on κ through Bayesian inference, making the activism premium liquidity-sensitive. Because the blockholder executes her trade at $P_{\text{trade}}(X)$ *before* the disclosure state $D = 1$ forces $\pi = 1$, she captures a stealth information rent. In the baseline calibration, this makes Public Voice viable for sufficiently strong signals despite the higher post-disclosure premium that bidders must pay.

4.8 Bid Incidence and Premia

The conditional bid probability $\tilde{p}(X, D)$ defined in (1) resolves the structural tension between sale facilitation and bid deterrence. Because $\Delta_S > 0$, an increase in inferred engagement strictly improves the core deal economics.

However, the bidder's required payout is anchored to the expected standalone value $\hat{V}(X, D)$, which increases by $\tilde{\Delta}$ for every unit of inferred engagement $\pi(X, D)$. Let $\Gamma(X, D)$ be the argument of the logistic CDF in (1). Differentiating the entry threshold yields:

$$\frac{\partial \tilde{p}(X, D)}{\partial \pi(X, D)} = -\Lambda' \left(\Gamma(X, D)\right) \frac{\tilde{\Delta} + (\tilde{m} - m_0) - \Delta_S}{s_\xi}.$$

Because the capitalized fundamental improvement and premium wedge exceed the synergy gain ($\tilde{\Delta} + \tilde{m} - m_0 > \Delta_S$ under Assumption (A5)), higher inferred engagement strictly *deters* bids ($\partial \tilde{p}/\partial \pi < 0$).

Crucially, this deterrence does not arise from frictional defensive tactics, but from *efficient market pricing*. The market capitalizes the activist's fundamental improvement into the standalone value, and the bidder must pay for it. When $D = 1$, the bidder *knows* activism has occurred, so the required fundamental compensation is maximal. Consequently, publicly disclosed activist campaigns deter marginal takeover bids more strongly than inferred activism. (Proof: see Appendix B.7.)

4.9 Equilibrium Cutoff Equations

The equilibrium cutoffs (k_1, k_0, k_D) are pinned down by indifference conditions at the boundaries of the blockholder's optimal action regions. Let U_E, $U_H(s)$, $U_Q(s)$, and $U_P(s)$ denote the expected payoffs from Exit, Hold, Quiet Voice, and Public Voice, computed by integrating over noise realizations and taking expectations of terminal payoffs. The expected payoff to exit, U_E, does not depend on the private signal s because the sell order $q = -1$ fully liquidates the position ($h = 0$). Her entire cash flow is determined by the anonymous execution price $P_{\text{trade}}(X)$, which is set competitively by the market maker based only on public order flow and equilibrium action probabilities, rendering it independent of the specific signal realization. In an interior equilibrium in which all four actions are played with positive probability, the three cutoffs satisfy adjacent indifference conditions:

$$U_E = U_H(k_1), \tag{7}$$

$$U_H(k_0) = U_Q(k_0), \tag{8}$$

$$U_Q(k_D) = U_P(k_D). \tag{9}$$

At k_1, the blockholder is indifferent between exiting and holding passively. At k_0, she is indifferent between Hold and Quiet Voice. At k_D, she is indifferent between Quiet Voice and Public Voice. If an intermediate action is globally dominated on the relevant signal range, its region collapses (e.g., if Quiet Voice is never optimal, then $k_0 = k_D$ and the boundary is given by $U_H(k_0) = U_P(k_0)$). As formally proven in Appendix B.2, the independence of $P_{\text{trade}}(X)$ from s structurally guarantees that the single-crossing property of the blockholder's payoff differences is preserved.

4.10 Equilibrium Existence and Uniqueness

Under Assumptions (A1)–(A7), any monotone-cutoff Perfect Bayesian Equilibrium must satisfy Bayes-consistent beliefs, the anonymous and post-disclosure pricing equations, and the cutoff indifference conditions (7)–(9).

Proposition 4 (Existence on a Compact Cutoff Domain)**.** *Suppose there exists a nonempty compact ordered cutoff domain $\Theta \subset \mathbb{R}^3$ such that the cutoff mapping $T : (k_1, k_0, k_D) \mapsto (k_1', k_0', k_D')$ is well defined on Θ, continuous on reached information sets with an arbitrary off-path completion, and maps Θ into itself. Then the model admits at least one monotone-cutoff Perfect Bayesian Equilibrium.*

Proof. See Appendix B.12.

The proof constructs the cutoff mapping $T : (k_1, k_0, k_D) \mapsto (k_1', k_0', k_D')$ and applies Brouwer's Fixed-Point Theorem on the maintained compact domain Θ. Because the post-disclosure price $P_{\text{post}}(X, D)$ is fully feed-forward, and the anonymous execution price $P_{\text{trade}}(X)$ is a finite convex combination of post-disclosure prices weighted by Bayes-consistent action probabilities, the main burden is to ensure that beliefs and cutoff updates remain well defined on reached states and extend continuously off path. I therefore present Proposition 4 as a regularity-based existence result rather than a fully global theorem.

Assumption (A6) is a numerical regularity condition on the cutoff mapping T. I do not claim an analytic uniqueness theorem. Instead, following the methodological precedent established for discrete order-flow feedback models (Edmans, Goldstein, et al., 2015), I report a computational fact: in all numerical exercises presented in this paper, iterating T from multiple starting points converges to the same fixed point. The paper therefore treats uniqueness as numerically verified rather than analytically proved.

4.11 Nonmonotonic Minority Takeover Gains

I now turn to the paper's central object of interest: expected minority takeover gains. These gains capture what a passive shareholder can expect to earn from the takeover channel, and the model predicts that they respond nonmonotonically to market liquidity. Define

$$\Delta^{\min}(\kappa) \equiv \mathbb{E}\big[m^R(a) \cdot \mathbf{1}\{\text{bid}\}\big], \tag{10}$$

as the expected takeover premium (per share) earned by a representative minority share. The expectation is taken over (v, s, z, ξ) induced by equilibrium strategies. This quantity captures the unconditional expected takeover premium from the takeover channel, averaged over all states of the world (including states in which no bid occurs).

Expected minority takeover gains decompose naturally into two components:

$$\Delta^{\min}(\kappa) = \underbrace{m_0 \cdot \mathbb{P}(\text{bid})}_{\text{baseline premium}} + \underbrace{(\tilde{m} - m_0) \cdot \mathbb{E}\big[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}\big]}_{\text{activism-driven premium}}. \tag{11}$$

The first term is the baseline premium weighted by overall bid probability. The second term, which I denote $\Delta^{\text{act}}(\kappa)$, captures the additional premium attributable to blockholder engagement. To avoid ambiguity, Δ (without superscript) denotes the engagement improvement parameter, while $\Delta^{\min}(\kappa)$ and $\Delta^{\text{act}}(\kappa)$ are equilibrium objects defined above.

This decomposition separates the effect of liquidity on *whether* a bid occurs from its effect on the *size* of the premium conditional on bidding. (Derivation: see Appendix B.11.)

Two opposing channels drive the liquidity dependence. Higher liquidity can raise bid incidence by reducing adverse-selection costs and shifting prices, but it also changes equilibrium engagement incentives and the inference-driven component of premia. A full closed-form characterization of $\Delta^{\min}(\kappa)$ is generally unavailable because κ affects equilibrium cutoffs and prices jointly. The next lemma describes the endpoint information structure; the overall shape of $\Delta^{\min}(\kappa)$ is then evaluated numerically.

Lemma 2 (Endpoint Behavior of Beliefs and Execution Prices with Bounded Noise)**.** *Fix any cutoff strategy (equivalently, fix the induced action probabilities* $(\omega_E, \omega_H, \omega_Q, \omega_P)$*).*

(i) *As* $\kappa \downarrow 0$*,* $\mathbb{P}(z = 0) \to 1$ *and hence* $X = q$ *almost surely. Therefore, on all reached order-flow states, the disclosure indicator* D *is (asymptotically) revealed by* X *and the anonymous execution price converges to the corresponding post-disclosure price:*

$$P_{\text{trade}}(X) \longrightarrow P_{\text{post}}(X, D(X)).$$

(ii) *As* $\kappa \uparrow 1$*, the noise distribution converges to* $\mathbb{P}(z = -1) = \mathbb{P}(z = 0) = \mathbb{P}(z = 1) = 1/3$*. Because the support of* z *is bounded, extreme order flows remain perfectly informative about disclosure (*$\mathbb{P}(D = 1 \mid X = 2) = 1$ *and* $\mathbb{P}(D = 1 \mid X \in \{-2, -1\}) = 0$*), while interior order flows (*$X \in \{0, 1\}$*) continue to mix across latent disclosure states via Bayes' rule.*

Proof. See Appendix B.14.

Proposition 5 (Baseline Numerical Nonmonotonicity)**.** *Minority takeover gains admit the decom-*

position

$$\begin{aligned}
\Delta^{\min}(\kappa) &= \Delta^{\text{base}}(\kappa) + \Delta^{\text{act}}(\kappa), \\
\Delta^{\text{base}}(\kappa) &\equiv m_0 \cdot \mathbb{P}(\text{bid}), \\
\Delta^{\text{act}}(\kappa) &\equiv (\tilde{m} - m_0) \cdot \mathbb{E}[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}] \,.
\end{aligned}$$

In the baseline calibration, $\Delta^{\text{base}}(\kappa)$ increases in liquidity κ while $\Delta^{\text{act}}(\kappa)$ decreases on the plotted grid, so that $\Delta^{\min}(\kappa)$ is hump-shaped with an interior maximizer. The sensitivity exercises in Section 5 illustrate how the peak location and even the presence of an interior maximizer can shift with primitives.

Proof. See Appendix B.15.

Remark (Shape). Continuity ensures that $\Delta^{\min}(\kappa)$ attains a maximum on a compact set, but whether that maximizer is interior is a quantitative matter. In the baseline calibration, and in the reported sensitivity exercises, the curve is single-peaked on the plotted range.

The hump arises because higher liquidity increases the incidence of takeovers (raising the baseline component) but reduces the extent to which activism is inferred and priced without disclosure (lowering the activism-driven component). With bounded noise trading, the right-endpoint $\kappa \uparrow 1$ does not generate a fully uninformative order-flow limit; instead, the decline in $\Delta^{\text{act}}(\kappa)$ is a general-equilibrium outcome driven by the endogenous reallocation of engagement across the quiet and public regimes, which we document numerically.

Section 5 complements this theory with numerical analysis. In the baseline calibration, $\Delta^{\min}(\kappa)$ is hump-shaped (Figure D.2) and the activism component varies smoothly with κ (Figure D.3).

Disclosure Attenuation. The activism-driven component $\Delta^{\text{act}}(\kappa)$ can be decomposed by disclosure status. In disclosed states ($D = 1$), engagement is known with certainty, so the activism premium $\tilde{m} - m_0$ is constant and there is no inference channel. In nondisclosed states ($D = 0$), engagement must be inferred from order flow, making the premium κ-sensitive through the Bayesian formulas in Proposition 2.

This implies that the liquidity sensitivity of $\Delta^{\text{act}}(\kappa)$ comes entirely from the $D = 0$ component. Shifting probability mass toward disclosed activism (larger ω_P) attenuates this inference-driven sensitivity, as more takeovers occur in states where engagement is observed rather than inferred.

Crucially, because the blockholder accumulates her stake at the anonymous execution price $P_{\text{trade}}(X)$ prior to the regulatory realization of D, she avoids the immediate adverse-selection penalty of the fully capitalized post-disclosure price $P_{\text{post}}(X, 1)$. In the baseline calibration, this informational stealth helps sustain Public Voice as a live high-signal action, so the attenuation result has genuine empirical bite.

Proposition 6 (Disclosure Attenuation (Partial Equilibrium))**.** *Hold the blockholder's strategy constant: fix the cutoffs (k_1, k_0, k_D) and hence the action probabilities, and vary κ only through its effect on inference and pricing (a partial equilibrium exercise). Decompose the activism-driven premium as*

$$\Delta^{\text{act}}(\kappa) = \underbrace{(\tilde{m} - m_0)\omega_P \cdot p(X, 1)}_{\textit{disclosed component}} + \underbrace{(\tilde{m} - m_0)\mathbb{E}\big[\pi(X, 0) \cdot p(X, 0) \mid D = 0\big]\mathbb{P}(D = 0)}_{\textit{inferred component}},$$

where $p(X,1)$ is the constant, invariant unconditional bid probability on the disclosed branch. Only the inferred component depends on κ through the posteriors $\pi(X,0)$. Consequently, $|\partial\Delta^{\text{act}}(\kappa)/\partial\kappa|$ is strictly decreasing in ω_P: shifting probability mass toward disclosed engagement attenuates the liquidity sensitivity of the activism-driven premium.

Proof sketch. In disclosed states, $\pi(X,1)=1$ and all post-disclosure pricing objects are κ-invariant (Appendix B.4), so the disclosed component contributes zero to $\partial\Delta^{\text{act}}/\partial\kappa$. The inferred component is a convex combination of κ-sensitive posteriors weighted by $\mathbb{P}(D=0)=1-\omega_P$. As ω_P increases, weight shifts from the κ-sensitive component to the κ-invariant component, reducing the overall sensitivity. □

Remark 2 (General Equilibrium). Proposition 6 is a partial-equilibrium statement. General-equilibrium disclosure effects are discussed heuristically in Section 8.4 and illustrated numerically; I do not claim a separate closed-form theorem there.

5 Comparative Statics and Numerical Analysis

This section combines analytical comparative statics with numerical analysis. Table C.3 in Appendix C summarizes the baseline parameterization; Table C.1 in Appendix C reports equilibrium outcomes.

5.1 Baseline Calibration and Equilibrium

The structural parameters (summarized in Table C.3) are calibrated to match the empirical realities of modern corporate governance and M&A markets. The baseline takeover probability is scaled by a Poisson arrival rate of $\lambda_B=0.20$, which, interacting with the logistic synergy dispersion ($s_\xi=0.15$) and baseline fundamental synergies ($\bar{S}=1.10$), strictly evaluates the target firm on the highly sensitive left tail of the entry threshold. This targets the unconditional annual M&A bid rate of 2% to 8% observed in standard U.S. firm-year panels (Betton et al., 2008), precluding the absurd base rates generated by unscaled frameworks. To capture the empirical reality that activists fundamentally force sales and dismantle entrenchment (Greenwood and Schor, 2009), successful engagement facilitates the sale by increasing expected synergy by $\Delta_S=0.30$.

Despite this synergy facilitation, the model rigorously satisfies the net deterrence condition (Assumption A5): the combined capitalization of the standalone fundamental improvement ($\tilde{\Delta}=0.225$) and the expected bargaining premium ($\tilde{m}-m_0=0.315$) strictly outpaces the synergy gain ($0.540>0.30$). Consequently, the marginal bidder faces a net reduction in deal surplus when acquiring a highly-priced, actively monitored firm. It is this perfectly rational price appreciation—not arbitrary defensive tactics—that cleanly deters the marginal bidder. Finally, the baseline engagement cost ($C_0=0.25$) is calibrated to exceed the expected fundamental improvement at the prior mean. This reflects the material frictions—legal fees, proxy solicitation, and illiquidity—that force blockholders with moderately positive signals to remain passive, safely reconstituting an interior *Hold* region. Figure D.1 in Appendix D depicts the resulting cutoff structure.

Three features of the equilibrium stand out:

- **Disclosed states have maximal activism premium:** $\bar{m}(X,1)=\tilde{m}$ for all $X\in\{0,1,2\}$, since $D=1$ reveals $a=1$ with certainty.

- **Nondisclosed states show inference:** The posterior $\pi(X, 0)$ is highest when order flow is most diagnostic of the quiet-engagement action. With no stake acquisition in $D = 0$ states, $X = 1$ reveals $q = 0$ and hence $\pi(1, 0) = \omega_Q/(\omega_H + \omega_Q)$.
- **Disclosure deters bids:** For fixed X, $p(X, 1) < p(X, 0)$ because the bidder knows it must pay the full activism premium when $D = 1$.

Figure D.4 plots equilibrium prices across all (X, D) states. Conditional on disclosure ($D = 1$), order flow reflects only noise given Public Voice, so $P_{\text{post}}(X, 1)$ is constant across X. The disclosed price level exceeds comparable nondisclosed states because the market knows engagement has occurred rather than inferring it.

5.2 Nonmonotonicity and Decomposition

Figure D.2 displays the central result: a nonmonotone relation between liquidity and expected minority takeover gains. As κ rises from low levels, $\Delta^{\min}(\kappa)$ initially increases because bid incidence grows. At higher liquidity, activism-related premia contribute less, and $\Delta^{\min}(\kappa)$ declines. The vertical dashed line marks the interior turning point $\kappa^{\dagger}$.

Figure D.3 decomposes these gains into the baseline component $m_0 \cdot \mathbb{P}(\text{bid})$ and the activism component $\Delta^{\text{act}}(\kappa)$. Both components vary smoothly with liquidity, and the hump shape arises from the interaction between bid incidence and the weight placed on activism-related premia.

Figure D.5 traces how equilibrium cutoffs shift with liquidity. The ordering $k_1 \leq k_0 \leq k_D$ is preserved throughout, though regions can collapse (e.g., $k_0 = k_1$).

5.3 Activism Frictions

Sensitivity to engagement frictions. The model's general equilibrium structure precludes simple closed forms, but analytical results pin down the direction of effects within each regime. Higher engagement costs shrink the voice regions; larger expected engagement benefits and takeover premia expand them.

Figures D.7 and D.8 illustrate how the nonmonotone pattern depends on key primitives.

Engagement cost C_0. Raising C_0 shrinks the voice regions and lowers the expected gains from engagement. In the numerical sweeps, low engagement costs can make $\Delta^{\min}(\kappa)$ increasing over the plotted range, while higher costs can generate an interior hump or shift the peak toward the boundary.

Premium wedge $(m_1 - m_0)$. Increasing the wedge raises the premium conditional on bidding but can deter bids through the bidder's entry condition. The net effect on $\Delta^{\min}(\kappa)$ is therefore ambiguous and can vary with κ.

5.4 Liquidity Effects

How does liquidity affect the market's inference about activism? Holding the blockholder's strategy fixed, the within-regime effects are as follows. Fix the equilibrium cutoffs (k_1, k_0, k_D) and hence the action probabilities $(\omega_E, \omega_H, \omega_Q, \omega_P)$. The posteriors in Proposition 2 satisfy:

(a) In nondisclosed states, $\partial\pi(1, 0)/\partial\kappa = 0$ and $\partial\pi(-1, 0)/\partial\kappa \geq 0$;

(b) $\partial\pi(0,0)/\partial\kappa \leq 0$;

(c) $\pi(-2,0) = 0$; in disclosed states, $\pi(X,1) = 1$ for all feasible X.

Consequently, the inferred premium wedge $\bar{m}(X,0) = m_0 + (\tilde{m} - m_0)\pi(X,0)$ is increasing in κ for $X = -1$, decreasing in κ for $X = 0$, and invariant for $X = 1$, while disclosed states have $\bar{m}(X,1) = \tilde{m}$. (Proof: see Appendix B.8.)

The intuition runs as follows. At $X = 1$, the buy order fully reveals that the blockholder did not sell, pinning down the posterior regardless of κ. At $X = -1$, higher liquidity makes a sell by the noise trader more likely, so the market revises upward the probability that the blockholder engaged quietly rather than exited. At $X = 0$, the opposite logic applies: higher liquidity makes the zero-order-flow event more likely under exit, reducing the inferred engagement probability.

Across-equilibrium liquidity effects. Beyond these within-regime inference effects, changing liquidity also shifts the blockholder's incentives through adverse selection and moves equilibrium cutoffs and prices jointly. Because these objects adjust endogenously with κ, the full cross-equilibrium patterns are characterized in Figure D.5.

In disclosed states ($D = 1$), $\pi(X,1) = 1$ and $\bar{m}(X,1) = \tilde{m}$, so liquidity has no inference content conditional on disclosure. In nondisclosed states ($D = 0$), $\pi(X,0)$ varies with κ through the Bayesian updating formulas in Proposition 2, making the $D = 0$ component of $\Delta^{\text{act}}(\kappa)$ liquidity-sensitive. This asymmetry between disclosed and nondisclosed states drives the model's core predictions.

Figure D.6 isolates the disclosure-inference channel. I fix the blockholder's cutoff strategy at the baseline equilibrium ($\kappa = 0.5$) and vary liquidity κ only through the noise distribution in order flow, then compare the resulting $\Delta^{\text{act}}(\kappa)$ under threshold disclosure to a counterfactual without disclosure. Removing disclosure amplifies the role of inference and makes $\Delta^{\text{act}}(\kappa)$ more sensitive to κ; threshold disclosure attenuates this sensitivity by shifting probability mass toward states with observed engagement ($D = 1$).

5.5 Takeover Environment

How do the takeover environment parameters shape bid incidence? Fix any reached information set (X, D) and treat the inferred premium wedge $\bar{m}(X, D)$ as given. The unconditional bid probability function $p(X, D)$ defined in (1) satisfies:

(a) **(Synergies)** $\partial p(X,D)/\partial\bar{S} > 0$: higher synergies increase bid incidence uniformly.

(b) **(Entry cost)** $\partial p(X,D)/\partial K < 0$: higher bidding costs reduce bid incidence uniformly.

Because the bid probability is anchored to the fundamental standalone value $\hat{V}(X, D)$ rather than the anticipatory market prices $P_{\text{trade}}(X)$ or $P_{\text{post}}(X, D)$, the price-to-entry feedback loop is structurally eliminated. The logistic synergy distribution with scale parameter s_ξ governs the dispersion of bidder valuations; larger s_ξ increases overall bid heterogeneity without affecting the feed-forward pricing structure.

(Proof: see Appendix B.9.)

5.6 Additional Sensitivity Analysis

I extend the sensitivity analysis to three additional structural parameters: the engagement success probability ρ, the bidder synergy scale s_ξ, and the discount factor δ.

First, I analyze the sensitivity to the engagement success probability $\rho \in \{0.88, 0.90, 0.92\}$ (Figure D.9). The baseline calibration utilizes $\rho = 0.9$, which is economically defensible because it represents the success rate conditional on the blockholder choosing to engage. Lowering ρ mechanically shrinks both the expected standalone improvement $\tilde{\Delta}$ and the expected premium wedge $\tilde{m} - m_0$. Numerically, this compresses the vertical scale of $\Delta^{\min}(\kappa)$ and can shift the location of its peak on the plotted range.

Second, I vary the bidder synergy scale $s_\xi \in \{0.10, 0.12, 0.15\}$ (Figure D.10). The parameter s_ξ governs the dispersion of the bidder's private logistic synergy shock. Increasing s_ξ changes bid incidence in a state-dependent way (it increases the chance of bidding in high-threshold states but moves probabilities toward 1/2 in low-threshold states). General-equilibrium cutoff adjustments further amplify these effects, so the resulting $\Delta^{\min}(\kappa)$ profile can flatten or shift substantially as s_ξ varies.

Finally, I test the sensitivity to the discount factor $\delta \in \{0.93, 0.95, 0.97\}$ (Figure D.11). The bidder's entry probability is anchored to fundamentals and inferred premia (Equation 1), so δ affects takeover outcomes only indirectly through equilibrium cutoffs and beliefs. Lower δ reduces the present value of terminal payoffs to the blockholder, weakening engagement incentives and shifting the cutoff vector (k_1, k_0, k_D); the numerical sweeps show that this can lower $\Delta^{\min}(\kappa)$ and alter the curvature pattern on the plotted range.

6 Testable Implications

The model generates several predictions regarding how liquidity, disclosure regimes, and takeover premia interact. Testing these structural predictions empirically requires navigating the extreme zero-inflation characteristic of unconditional takeover gains. To avoid Type II errors common to causal quasi-experiments on zero-inflated panels, the empirical strategy must isolate the extensive and intensive margins directly, utilizing highly powered tests of curvature.

1. **The Unconditional Hump Test.** In the baseline calibration, extreme illiquidity suppresses bids via efficient pricing, while at high liquidity the activism-driven premium component is eroded by diluted inference and the equilibrium reallocation of engagement toward disclosed states, yielding a nonmonotone maximum for minority gains (Proposition 5). *Prediction:* The relationship between market liquidity and unconditional expected minority takeover gains can display an inverted-U shape. *Empirical strategy:* Estimate a quadratic specification of $Y_{i,t} = \text{Premium}_{i,t} \times \mathbf{1}\{\text{Bid}_{i,t} = 1\}$ on $L_{i,t}$ and $L_{i,t}^2$ (where $Y = 0$ for non-targets), controlling for institutional ownership to separate active noise from passive indexing. Formally test for the inverted-U using the exact Lind and Mehlum (2010) joint slope test. To address nonparametric functional-form concerns, this curvature can be cross-validated using Double Machine Learning (DML).

2. **The Reduced-Form Decomposition.** Equation (11) analytically separates takeover gains into a baseline bid-incidence component and an activism-driven premium component. In the

baseline calibration, the inverted-U pattern comes from these two forces moving in opposite directions. *Prediction:* Where the unconditional hump appears, the extensive margin (bid incidence) rises with liquidity while the intensive-margin activism premium declines as order flow loses informativeness. *Empirical strategy:* Decompose the effect. First, estimate a Probit model on the hazard rate of receiving an M&A bid to capture the extensive margin. Second, estimate an OLS regression purely on the conditional premium for completed deals to confirm the intensive margin does not mechanically offset the unconditional pattern.

3. **Mechanism Validation via Observability Interaction.** The inference channel strictly operates in the nondisclosed state ($D = 0$). Shifting probability mass from inferred to observable states inherently attenuates the sensitivity of the premium to liquidity (Proposition 6). *Prediction:* Higher public observability of the target firm attenuates the curvature of the liquidity-premia hump. *Empirical strategy:* Interact the quadratic liquidity terms with continuous proxies for the target's information environment (e.g., analyst coverage, index inclusion). The model predicts that for highly covered firms, the quadratic coefficient on L^2 will pull significantly toward zero, mathematically flattening the hump because engagement is priced directly rather than inferred from noisy order flow.

4. **Micro-Validation via 13D Event Studies.** In the baseline action set, a Schedule 13D filing maps the market from inferred activism to observed public intervention, so the filing-day price jump directly measures the gap between pre-filing inference $\pi(X, 0)$ and the disclosed state. *Prediction:* 13D cumulative abnormal returns (CARs) are strictly larger for highly liquid stocks. *Empirical strategy:* Regress target CARs around Schedule 13D filings on pre-event Amihud illiquidity. In highly liquid stocks (high κ), order flow is less informative about hidden engagement, so pre-filing beliefs are more diffuse and the filing represents a larger informational shock, yielding larger CARs. Conversely, in illiquid stocks, order flow is more revealing, inference occurs earlier through execution prices, and the filing-day surprise is smaller.

7 Preliminary Welfare Accounting

The analysis thus far has focused on expected minority takeover gains, $\Delta^{\min}(\kappa)$, which represent the natural welfare object for dispersed shareholders facing a free-rider problem. A broader perspective requires analyzing total economic surplus. Because this accounting is sensitive to how bidder rents, takeover transfers, and activism costs are normalized, I treat this section as suggestive rather than definitive. Define the welfare components as follows: minority welfare $W_{\min}(\kappa) = \Delta^{\min}(\kappa)$; blockholder welfare $W_B(\kappa) = \mathbb{E}[U(q^*, a^* \mid s)]$; bidder welfare $W_{\text{bid}}(\kappa) = \mathbb{E}[\max(\Pi_B, 0)]$; and total surplus $W(\kappa) = W_{\min} + W_B + W_{\text{bid}}$.

In the event of a takeover, the expected standalone value $\hat{V}(X, D)$ and the premium wedge $m^R(a)$ act as pure transfers from the bidder to the target shareholders. Consequently, these pricing terms cancel out in the aggregate. Expected total surplus simplifies to the realized synergies and standalone improvements, net of costs:

$$W(\kappa) = \mathbb{E}\Big[\mathbf{1}\{\text{bid}\}(\bar{S} + \xi - K - v - a\tilde{\Delta}) + v + a\tilde{\Delta} - aC(s)\Big]. \tag{12}$$

An important question is whether the nonmonotonicity of minority gains is merely a distributional artifact or also a feature of total surplus. Because takeover premia are transfers, the total-surplus criterion depends on realized synergies and standalone improvements net of engagement costs, not on how the surplus is split. In the baseline calibration, total surplus $W(\kappa)$ is increasing over the plotted liquidity range (Figure D.13), even though minority gains are hump-shaped.

Consequently, the socially optimal liquidity level κ^* can differ from the minority-optimal level $\kappa^\dagger$ in the baseline calibration. Intuitively, higher liquidity reduces adverse-selection costs and raises bid incidence, improving bidder surplus and aggregate surplus, whereas minority takeover gains peak earlier because they load on the inference-driven activism premium, which is eroded as order flow becomes less informative and engagement becomes more observable. I therefore view Figure D.13 as a preliminary accounting exercise rather than a policy theorem.

8 Extensions

Disclosure requirements for activist investors vary widely across jurisdictions, creating natural laboratories for testing the model's predictions. I consider three benchmarks that span the policy spectrum.

In the United States, Schedule 13D mandates filing within five business days of crossing a 5% beneficial ownership threshold. The United Kingdom imposes a lower 3% threshold with a tighter two-trading-day deadline under the FCA's Disclosure and Transparency Rules. The European Union's Transparency Directive sets a 5% baseline but permits member states to adopt lower thresholds; Germany, for instance, applies 3% for voting-rights notifications. At one extreme, very low thresholds approach full disclosure of activist intent; at the other, weak enforcement or broad institutional exemptions leave much activism unobserved.

Throughout this section, I hold the blockholder's cutoff strategy fixed, and hence the action probabilities $\omega_E, \omega_H, \omega_Q, \omega_P$, to isolate the effect of the disclosure regime on inference and pricing.

8.1 Full Disclosure Benchmark

Some jurisdictions have considered or implemented very low disclosure thresholds or require disclosure of activist intent regardless of stake size. In the limit, such regimes approach full disclosure.

Under full disclosure, the market observes the blockholder's action (q, a) directly at $t = 1.2$. The engagement posterior is trivial: $\pi = 1$ when $a = 1$ and $\pi = 0$ when $a = 0$. The activism premium is deterministic in each state, and the inference channel that links liquidity to premia shuts down completely. The pre-disclosure anonymous execution price $P_{\text{trade}}(X)$ converges to the expectation over these perfectly separated terminal states. This benchmark marks the upper bound on how much disclosure can attenuate the liquidity-sensitivity of the activism premium.

8.2 No Disclosure Benchmark

At the other extreme, some jurisdictions have high thresholds, weak enforcement, or broad exemptions that leave much activism unobserved. The no-disclosure benchmark captures this limiting case: the market receives no regulatory disclosure signal D and must infer everything from order flow alone.

Under no disclosure, the $t = 1.2$ regulatory update is entirely absent. The market conditions solely on aggregate order flow $X \in \{-2, -1, 0, 1, 2\}$. Consequently, the anonymous execution price equals the secondary-market price (there is no post-filing jump): $P_{\text{trade}}(X) = P_{\text{post}}(X)$. The blockholder therefore fully internalizes the adverse-selection cost of her trade. The market mixes over all four actions with weights that depend on the noise distribution and hence on κ. Compared to the baseline, this regime forces the activism premium to be highly liquidity-sensitive across all states.

8.3 An Intermediate Regime: Stake Disclosure Plus Noisy Rumors

Between the polar benchmarks of full disclosure and no disclosure lies a realistic intermediate case: baseline stake-triggered disclosure augmented by a noisy public signal about quiet engagement. This regime captures modern information environments, such as the United Kingdom, where a lower 3% threshold is augmented by active financial media (L. Fang and Peress, 2009).

To formalize this, I augment the baseline model with a binary public rumor signal $R \in \{0, 1\}$ that fires in nondisclosed states ($D = 0$) alongside the $t = 1.2$ regulatory update. If the blockholder engages ($a = 1$), the rumor fires with true-positive probability η_1. If she does not engage ($a = 0$), the rumor fires with false-positive probability η_0, where $\eta_0 < \eta_1$.

Applying Bayes' rule, the post-disclosure updated posterior in the nondisclosed state becomes:

$$\pi(X, 0, R) = \frac{\omega_Q \mathbb{P}(X|q = 0)\mathbb{P}(R|a = 1)}{\omega_E \mathbb{P}(X|q = -1)\mathbb{P}(R|a = 0) + \mathbb{P}(X|q = 0)(\omega_H \mathbb{P}(R|a = 0) + \omega_Q \mathbb{P}(R|a = 1))}. \tag{13}$$

When the rumor fires ($R = 1$), the market updates toward engagement; when it remains silent ($R = 0$), the market updates away. The anonymous execution price $P_{\text{trade}}(X)$ integrates over both the latent disclosure D and the latent rumor outcome R.

Numerical analysis of this regime (Figure D.12) reveals that as the rumor becomes more precise (i.e., as $\eta_1 - \eta_0$ increases), the $\Delta^{\min}(\kappa)$ curve systematically flattens.

8.4 A Heuristic General-Equilibrium Disclosure Trade-off

Proposition 6 established a partial-equilibrium result: holding the blockholder's strategy fixed, stricter disclosure attenuates the liquidity sensitivity of premia by increasing transparency. However, in general equilibrium, altering the regulatory disclosure threshold fundamentally changes the blockholder's incentive to engage in the first place.

Let τ parameterize the strictness of the disclosure regime. The total derivative of the activism-driven premium with respect to disclosure strictness can be conceptually decomposed as:[7]

$$\frac{d\Delta^{\text{act}}}{d\tau} \approx \underbrace{\frac{\partial \Delta^{\text{act}}}{\partial \tau}}_{\text{Transparency Effect}} + \underbrace{\frac{\partial \Delta^{\text{act}}}{\partial(\omega_Q + \omega_P)} \frac{d(\omega_Q + \omega_P)}{d\tau}}_{\text{Deterrence Effect}}. \tag{14}$$

The transparency effect is positive in the heuristic decomposition: shifting mass to observable states

[7] This decomposition serves as an intuitive heuristic for the economic logic. Formally, Δ^{act} is a highly nonlinear function of all individual region probabilities (ω_E, ω_H, ω_Q, ω_P), and shifting mass between distinct regions entails asymmetric effects on the Bayesian posteriors that require a full multivariate application of the chain rule.

($D = 1$) removes the inference discount. The deterrence effect is negative: stricter thresholds force early disclosure, which can reduce the expected return to quiet engagement and shrink the total probability of activism.

The net impact on minority shareholders is therefore ambiguous and depends on primitives. Under sufficiently low engagement costs, the transparency effect can dominate. When engagement is already marginal, stricter disclosure can instead deter the very activism that generates takeover premia. I view this subsection as economic guidance for future work rather than a proved comparative static.

9 Conclusion

This paper develops a unified theoretical framework in which secondary-market liquidity, activism disclosure, and takeover premia are jointly determined in equilibrium. By embedding an informed blockholder's choice between Exit, Quiet Voice, and Public Voice into a setting with order-flow inference and takeover feedback, the model isolates an activism premium that is either directly observed through regulatory disclosure or inferred by a competitive market maker. The main analytical contribution is a tractable decomposition of beliefs, prices, and takeover premia across disclosed and nondisclosed states. The main quantitative finding is that, in the baseline calibration, liquidity exerts a nonmonotone effect on expected minority takeover gains. Rising noise-trading intensity lowers adverse-selection costs and spurs bid incidence, but simultaneously erodes the informativeness of order flow, dampening the inference channel that sustains activism-related premia.

The analysis provides several distinct, testable implications. Most notably, threshold-based disclosure policy acts as an implicit governor on the liquidity-premia relationship. Lowering the disclosure threshold increases the share of activism that is directly observable, thereby attenuating the inference channel. Consequently, the model predicts that the sensitivity of takeover premia to market liquidity is structurally lower in strict-disclosure jurisdictions (such as the United Kingdom) than in permissive jurisdictions (such as the United States). The more ambitious general-equilibrium disclosure and welfare questions remain open, but the current framework already clarifies how disclosure and public-market inference interact in takeover settings.

Extending the framework to dynamic settings with multiple trading rounds would capture the gradual position-building that characterizes real-world activist campaigns. Introducing multiple blockholders or endogenous information acquisition would illuminate how collective action and rumor networks, such as the wolf pack dynamics explored in Section 8.3, reshape the inference channel. Each of these extensions preserves the core insight developed here: liquidity, disclosure, and corporate governance are inextricably linked, and evaluating any one requires a framework that internalizes their joint equilibrium feedback.

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A Notation

Symbol	Meaning
Fundamentals and information	
v	Standalone fundamental at $t = 0$, $v \sim \mathcal{N}(\mu, \sigma_v^2)$.
s	Blockholder signal, $s = v + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
$\mu, \sigma_v^2, \sigma_\varepsilon^2, \sigma_s^2$	Prior mean/variances; $\sigma_s^2 \equiv \sigma_v^2 + \sigma_\varepsilon^2$.
β	Posterior weight in $\hat{v}(s) = \mathbb{E}[v \mid s] = \mu + \beta(s - \mu)$; $\beta \equiv \sigma_v^2/(\sigma_v^2 + \sigma_\varepsilon^2)$.
$\hat{v}(s)$	Blockholder posterior mean of v given s.
Φ, ϕ	Standard normal CDF and PDF.
$\lambda_L(\alpha), \lambda_U(\alpha)$	Inverse Mills ratios: $\lambda_L(\alpha) = \phi(\alpha)/\Phi(\alpha)$ and $\lambda_U(\alpha) = \phi(\alpha)/(1 - \Phi(\alpha))$.
Trading, liquidity, disclosure	
q	Blockholder order at $t = 1$, $q \in \{-1, 0, +1\}$.
z	Noise-trader order, $z \in \{-1, 0, +1\}$ with $\mathbb{P}(z = 0) = 1 - \frac{2}{3}\kappa$ and $\mathbb{P}(z = \pm 1) = \frac{\kappa}{3}$.
κ	Noise-trading intensity (liquidity).
X	Total order flow observed by the market maker, $X = q + z$.
D	Disclosure indicator for threshold-crossing stake acquisition, $D = 1 \iff (q = +1)$.
$P_{\text{trade}}(X)$	Anonymous execution price at $t = 1$ conditional on order flow X.
$P_{\text{post}}(X, D)$	Post-disclosure secondary market price conditional on (X, D).
Engagement (voice)	
a	Engagement choice at $t = 1$, $a \in \{0, 1\}$.
$C(s), C_0, \chi$	Engagement cost at $t = 1$ (private to the blockholder); $C(s) = C_0 \exp\left(-\chi \frac{s-\mu}{\sigma_s}\right)$.
ρ	Engagement success probability.
$\Delta, \tilde{\Delta}$	Standalone improvement if engagement succeeds; $\tilde{\Delta} \equiv \rho\Delta$ is the expected improvement under $a = 1$.
m_0, m_1	Takeover premia above $\hat{V}$: baseline m_0 (no successful engagement) and $m_1 > m_0$ (successful engagement).
$\tilde{m}$	Expected premium under engagement, $\tilde{m} \equiv m_0 + \rho(m_1 - m_0)$.
$\pi(X, D)$	Posterior engagement probability, $\pi(X, D) \equiv \mathbb{P}(a = 1 \mid X, D)$.
$\bar{m}(X, D)$	Expected premium wedge conditional on (X, D), $\bar{m}(X, D) = \mathbb{E}[m^R(a) \mid X, D] = m_0 + (\tilde{m} - m_0)\pi(X, D)$.
Takeover and payoffs	
$\bar{S}$	Baseline bidder synergy (incremental value from acquisition).
Δ_S	Expected fundamental synergy improvement from activism.
K	Fixed bidding / entry cost.
ξ, s_ξ	Bidder's private synergy shock, $\xi \sim \Lambda(0, s_\xi)$ (logistic distribution).
λ_B	Exogenous Poisson arrival probability of a potential bidder.
$\Pi_B(X, D)$	Bidder expected net deal surplus conditional on (X, D).
$\tilde{p}(X, D)$	Conditional bid probability given bidder arrival.
$p(X, D)$	Unconditional bid probability conditional on (X, D); $p(X, D) = \lambda_B \cdot \tilde{p}(X, D)$.
$m^R(a)$	Realized premium wedge conditional on engagement: $m^R(a) = m_0 + a(\tilde{m} - m_0)$.
$b(X, D)$	Expected per-share offer if a bid occurs, $b(X, D) = \hat{V}(X, D) + \bar{m}(X, D)$.
τ	Time between pricing/entry and settlement in $\delta = \exp(-r\tau)$.
δ	Discount factor between pricing/entry and settlement, $\delta = \exp(-r\tau)$ (or $1/(1 + r)$).
Y	Terminal per-share payoff (takeover vs. standalone).
$U(q, a \mid s)$	Blockholder expected utility at $t = 1$ conditional on s and action (q, a).
$\Delta^{\min}(\kappa)$	Expected takeover premium earned by a representative minority share, $\mathbb{E}[m^R(a) \cdot \mathbf{1}\{\text{bid}\}]$.
$\Delta^{\text{act}}(\kappa)$	Activism-driven component of minority takeover gains, $\mathbb{E}[(m^R(a) - m_0) \cdot \mathbf{1}\{\text{bid}\}]$.
Equilibrium thresholds and regions	
k_1, k_0, k_D	Signal cutoffs for Exit, Hold, Quiet Voice (hidden engagement), and Public Voice (disclosed engagement).
α_i	Standardized cutoffs $\alpha_i \equiv (k_i - \mu)/\sigma_s$ for $i \in \{1, 0, D\}$.
$\omega_E, \omega_H, \omega_Q, \omega_P$	Ex ante probabilities of Exit / Hold / Quiet Voice / Public Voice regions.

Table 1: Notation summary (quick reference).

Label	Content	Location
(A1)	Interior cutoffs (nondegeneracy)	Engagement Technology (§3)
(A2)	Engagement cost decreasing in signal	Engagement Technology (§3)
(A3)	Premium wedge: $\tilde{m} > m_0$	Engagement Technology (§3)
(A4)	Interior bid probability	Bidder Entry (§3)
(A5)	Net deterrence condition: $\tilde{\Delta} + \tilde{m} - m_0 > \Delta_S$	Bidder Entry (§3)
(A6)	Numerical regularity / stable fixed-point selection for T	Equilibrium Concept (§3)
(A7)	Bounded bidder arrival rate: $\lambda_B \leq 1/2$	Bidder Entry (§3)

Table 2: Summary of standing assumptions (A1)–(A7).

B Proofs and Derivations

B.1 Proof of Lemma 1 (Domination of Passive Accumulation)

Proof. Let $QA \equiv (+1, 0)$ denote Quiet Accumulation and let $P \equiv (+1, 1)$ denote Public Voice. Fix an arbitrary belief system $\pi(X, 1) \in [0, 1]$ upon observing $D = 1$ and consider any signal realization s.

Because $q = +1$ under both QA and P, the aggregate order flow is $X = 1+z$ and the anonymous execution price $P_{\text{trade}}(X)$ applies identically under both actions. Moreover, conditional on $(X, 1)$, the bid probability $p(X, 1)$ and the post-disclosure price $P_{\text{post}}(X, 1)$ are identical under QA and P because a is not observed at the trading stage. Thus the only difference between QA and P is the realized engagement decision $a \in \{0, 1\}$, which changes (i) the realized premium from m_0 to $\tilde{m}$ when a bid occurs and (ii) the standalone payoff from v to $v + \tilde{\Delta}$ when no bid occurs, at private cost $C(s)$.

Write $\alpha \equiv \tilde{m} - m_0 > 0$ and $\beta \equiv \tilde{\Delta} > 0$. Using the payoff function (15) and the fact that $h = 2$ shares are held under both QA and P, we have:

$$\begin{aligned} U_P(s) = \mathbb{E}_z\Big[&- P_{\text{trade}}(1+z) \\ &+ 2\delta\big(p(1+z, 1)\,(\hat{V}(1+z, 1) + \tilde{m}) \\ &\qquad + (1 - p(1+z, 1))(\hat{v}(s) + \tilde{\Delta})\big)\Big] - C(s), \\ U_{QA}(s) = \mathbb{E}_z\Big[&- P_{\text{trade}}(1+z) \\ &+ 2\delta\big(p(1+z, 1)\,(\hat{V}(1+z, 1) + m_0) \\ &\qquad + (1 - p(1+z, 1))\hat{v}(s)\big)\Big]. \end{aligned}$$

Subtracting and cancelling the common execution-price and $\hat{V}(1 + z, 1)$ terms yields:

$$U_P(s) - U_{QA}(s) = 2\delta\, \mathbb{E}_z\Big[p(1+z, 1)\alpha + (1 - p(1+z, 1))\beta\Big] - C(s).$$

For any $p \in [0, 1]$, the expression $p\alpha + (1 - p)\beta$ is a convex combination of α and β, hence satisfies

$$p\alpha + (1 - p)\beta \;\geq\; \min\{\alpha, \beta\}.$$

Applying this pointwise inequality inside the expectation gives

$$U_P(s) - U_{QA}(s) \;\geq\; 2\delta \min\{\alpha, \beta\} - C(s).$$

By Assumption (A2), $C(s)$ is strictly decreasing and $\lim_{s\to\infty} C(s) = 0$. Define the positive constant $\underline{G} \equiv 2\delta \min\{\alpha, \beta\} > 0$. Since $C(s) \to 0$, there exists a finite $\bar{s}$ such that $C(s) < \underline{G}$ for all $s \geq \bar{s}$. For such s,

$$U_P(s) - U_{QA}(s) \;\geq\; \underline{G} - C(s) \;>\; 0,$$

so $U_P(s) > U_{QA}(s)$ for all $s \geq \bar{s}$, *regardless of the market's off-path belief* $\pi(X, 1)$. □

Remark 1 (Strengthening via an additional assumption)**.** *The lemma as stated guarantees domination only for sufficiently high signals $s \geq \bar{s}$. If a stronger "for all $s \geq k_0$" conclusion is desired, it suffices to add an assumption ensuring the engagement cost at the bottom of the voice region is below the belief-free gross benefit:*

$$\text{(A8)} \qquad C(k_0) \;<\; 2\delta \min\{\tilde{m} - m_0,\; \tilde{\Delta}\}.$$

Because $C(s)$ is decreasing, $s \geq k_0$ implies $C(s) \leq C(k_0) < \underline{G}$, so $U_P(s) - U_{QA}(s) > 0$ for all $s \geq k_0$. In the baseline calibration, (A8) is satisfied with substantial margin.

B.2 Proof of Proposition 1

Proof. Write the feasible actions as

$$E \equiv (-1, 0), \quad H \equiv (0, 0), \quad Q \equiv (0, 1), \quad P \equiv (+1, 1),$$

corresponding to Exit, Hold, Quiet Voice, and Public Voice. This is the baseline action set used in the equilibrium definition; Lemma 1 provides supporting intuition for why passive threshold crossing is omitted.

Step 1: Best responses are characterized by (at most) three cutoffs. Fix any candidate post-disclosure pricing rule $P_{\text{post}}(\cdot, \cdot)$, anonymous execution rule $P_{\text{trade}}(\cdot)$, and bidder-entry probabilities $p(\cdot, \cdot)$. For an action (q, a) with holdings $h = 1 + q$ and disclosure $D = \mathbf{1}\{q = +1\}$, the blockholder's $t = 1$ expected payoff conditional on signal s is

$$U(q, a \mid s) = \mathbb{E}\Big[-q \cdot P_{\text{trade}}(X) + \delta h \cdot Y - a \cdot C(s) \,\Big|\, s, q, a \Big],$$

where $X = q + z$ and $Y = \mathbf{1}\{\text{bid}\} \cdot b(X, D, a) + (1 - \mathbf{1}\{\text{bid}\}) \cdot (v + a\tilde{\Delta})$. Since the bid event depends only on ξ conditional on (X, D), $\mathbb{E}[\mathbf{1}\{\text{bid}\} \mid X, D] = p(X, D)$, and risk neutrality implies $\mathbb{E}[v \mid s] = \hat{v}(s)$. Taking expectations over (v, ξ, z) conditional on s yields

$$\begin{aligned} U(q, a \mid s) = \mathbb{E}_z\Big[-q \cdot P_{\text{trade}}(X) + \delta h\big(p(X, D) \cdot (\hat{V}(X, D) + m^R(a)) \\ + (1 - p(X, D)) \cdot (\hat{v}(s) + a\tilde{\Delta})\big)\Big] - a \cdot C(s). \end{aligned}$$

Because the anonymous execution price $P_{\text{trade}}(X)$ is determined by the market maker's unconditional expectation over aggregate order flow prior to disclosure, it is strictly independent of the blockholder's private signal s. Define the signal-independent slope coefficient:

$$B_{q,a} \equiv \delta h \cdot \mathbb{E}_z[1 - p(X, D)], \qquad X = q + z,$$

and collect all remaining terms (including the s-invariant execution cash flow $\mathbb{E}_z[-qP_{\text{trade}}(X)]$) into a constant $A_{q,a}$:

$$A_{q,a} \equiv \mathbb{E}_z\Big[- qP_{\text{trade}}(X) + \delta h\big(p(X, D)(\hat{V}(X, D) + m^R(a)) + (1 - p(X, D))a\tilde{\Delta}\big)\Big].$$

Then the blockholder's utility is strictly affine in the posterior fundamental $\hat{v}(s)$:

$$U(q, a \mid s) = A_{q,a} + B_{q,a}\, \hat{v}(s) - a \cdot C(s). \tag{15}$$

Because $\hat{v}(s) = \mu + \beta(s - \mu)$ is strictly increasing in s, and $C(s)$ is weakly decreasing by Assumption (A2), the derivative $\frac{\partial U}{\partial s} = B_{q,a}\beta - aC'(s)$ is strictly positive. Crucially, the anonymous execution price $P_{\text{trade}}(X)$ vanishes entirely from the derivative. The payoff differences that determine the adjacent-action indifference cutoffs satisfy the strict single-crossing property in s. In particular:

- **Exit vs. Hold.** $U_H(s) - U_E = A_H - A_E + B_H\hat{v}(s)$ is strictly increasing in s (since $B_H > 0$ for $h = 1$). Hence there is at most one cutoff k_1 solving $U_E = U_H(k_1)$.
- **Hold vs. Quiet Voice.** Since H and Q share the exact same trading behavior $q = 0$, the execution cash flow $P_{\text{trade}}(X)$ perfectly cancels out from $A_{q,a}$. They also share $D = 0$ and $h = 1$, so their $B_{q,a}$ coefficients are identical. The difference reduces exactly to:

$$U_Q(s) - U_H(s) = \delta\, \mathbb{E}_z\Big[p(X, 0) \cdot (\tilde{m} - m_0) + (1 - p(X, 0)) \cdot \tilde{\Delta}\Big] - C(s), \tag{16}$$

  which is strictly increasing in s when $\chi > 0$. Thus there is at most one cutoff k_0 solving $U_H(k_0) = U_Q(k_0)$.
- **Quiet vs. Public Voice.** Since Q and P both have $a = 1$, the cost term $C(s)$ cancels in the difference. We group the terms involving $\hat{v}(s)$ to show $U_P(s) - U_Q(s)$ is an affine function. The coefficient on s is $B_P = 2\delta\beta\mathbb{E}_z[1 - p(X, 1)]$ for Public Voice and $B_Q = \delta\beta\mathbb{E}_z[1 - p(X, 0)]$ for Quiet Voice. The difference in slopes evaluates to:

$$B_P - B_Q = \delta\beta\, \mathbb{E}_z\Big[2(1 - p(X, 1)) - (1 - p(X, 0))\Big] = \delta\beta\, \mathbb{E}_z\Big[1 - 2p(X, 1) + p(X, 0)\Big].$$

  Because the unconditional bid probability is structurally bounded by the bidder arrival rate ($p(X, D) \leq \lambda_B$), Assumption (A7) implies $2p(X, 1) \leq 1$. In nondegenerate calibrations with $p(X, 1) < 1/2$, the term inside the expectation is strictly positive, so $\frac{\partial}{\partial s}(U_P - U_Q) = \beta(B_P - B_Q) > 0$. The marginal fundamental benefit of retaining a second share is therefore strictly increasing in the signal s, yielding at most one cutoff k_D solving $U_Q(k_D) = U_P(k_D)$.

Under Assumption (A1) (interiority/nondegeneracy of regions), the relevant indifference cutoffs exist and are (weakly) ordered, yielding a best response with thresholds $k_1 \leq k_0 \leq k_D$.

Step 2: Given cutoffs, beliefs and prices are pinned down. Fix a cutoff vector (k_1, k_0, k_D). This pins down the ex ante region probabilities $(\omega_E, \omega_H, \omega_Q, \omega_P)$ and hence (by Bayes' rule) the posteriors $\pi(X, D)$ and the latent disclosure probabilities $\mathbb{P}(D = d \mid X)$. Given these posteriors and conditional means $\mathbb{E}[v \mid X, D]$, the expected standalone value $\hat{V}(X, D)$ and unconditional bid probability $p(X, D)$ are strictly determined. The post-disclosure competitive pricing rule is directly defined by the explicit feed-forward equation (5) to yield $P_{\text{post}}(X, D)$, which linearly folds back to yield the anonymous $P_{\text{trade}}(X)$ via (6).

Step 3: Fixed point over cutoff vectors. Define the cutoff mapping $T : (k_1, k_0, k_D) \mapsto (k_1', k_0', k_D')$ by: (i) compute $(\omega_E, \omega_H, \omega_Q, \omega_P)$; (ii) compute posteriors and $\mathbb{P}(D = d|X)$; (iii) directly compute feed-forward prices P_{post} and P_{trade}; and (iv) define (k_1', k_0', k_D') as the (unique) solutions to the indifference conditions (7)–(9) induced by these objects. By construction, each step is continuous, so T is continuous.

If T maps a compact ordered cutoff domain into itself and is continuous there, Brouwer's theorem gives at least one fixed point. If, in addition, Assumption (A6) holds, the numerical solver selects a unique fixed point by convergence from multiple starting values. At any such fixed point, the blockholder's strategy is optimal given the induced prices, beliefs are Bayes-consistent, and the market maker's dual pricing rules are competitive, constituting a Perfect Bayesian Equilibrium. □

B.3 Proof of Proposition 2

Proof. For disclosed states, $D = 1$ occurs if and only if $q = +1$ by the disclosure rule. Since the only feasible action with $q = +1$ is Public Voice, $\pi(X, 1) = \mathbb{P}(a = 1 \mid X, 1) = 1$ for all feasible $X \in \{0, 1, 2\}$.

For nondisclosed states, $D = 0$ rules out Public Voice. The remaining actions consistent with $D = 0$ are Exit $E \equiv (-1, 0)$, Hold $H \equiv (0, 0)$, and Quiet Voice $Q \equiv (0, 1)$, which occur with ex ante probabilities $(\omega_E, \omega_H, \omega_Q)$.

Write the noise trade probabilities as $\mathbb{P}(z = 0) = 1 - \frac{2}{3}\kappa$ and $\mathbb{P}(z = \pm 1) = \frac{\kappa}{3}$. Conditional on each action, $X = q + z$ has the following support and probabilities:

	$X = q - 1$	$X = q$	$X = q + 1$
$q = -1$ (E)	$\kappa/3$	$1 - \frac{2}{3}\kappa$	$\kappa/3$
$q = 0$ (H, Q)	$\kappa/3$	$1 - \frac{2}{3}\kappa$	$\kappa/3$
$q = +1$ (P)	$\kappa/3$	$1 - \frac{2}{3}\kappa$	$\kappa/3$

Note that Hold H and Quiet Voice Q generate identical order-flow distributions since both have $q = 0$. Applying Bayes' rule, for any X with $D = 0$,

$$\pi(X, 0) = \mathbb{P}(a = 1 \mid X, 0) = \mathbb{P}(Q \mid X, 0) = \frac{\omega_Q \, \mathbb{P}(X \mid Q)}{\omega_E \, \mathbb{P}(X \mid E) + (\omega_H + \omega_Q) \, \mathbb{P}(X \mid q{=}0)}.$$

If $X = -2$, then X can only be generated by Exit, so $\pi(-2, 0) = 0$.

If $X = 1$, then $X = 1$ requires $q = 0$ and $z = +1$, so conditioning on $D = 0$ implies

$$\pi(1, 0) = \frac{\omega_Q}{\omega_H + \omega_Q}.$$

If $X = -1$, then $\mathbb{P}(X = -1 \mid q{=}0) = \mathbb{P}(z = -1) = \frac{\kappa}{3}$ and $\mathbb{P}(X = -1 \mid E) = \mathbb{P}(z = 0) = 1 - \frac{2}{3}\kappa$, giving

$$\pi(-1,0) = \frac{\omega_Q(\kappa/3)}{(\omega_H+\omega_Q)(\kappa/3)+\omega_E(1-\frac{2}{3}\kappa)}.$$

If $X=0$, then $\mathbb{P}(X=0 \mid q{=}0) = \mathbb{P}(z=0) = 1-\frac{2}{3}\kappa$ and $\mathbb{P}(X=0 \mid E) = \mathbb{P}(z=1) = \frac{\kappa}{3}$, so

$$\pi(0,0) = \frac{\omega_Q(1-\frac{2}{3}\kappa)}{(\omega_H+\omega_Q)(1-\frac{2}{3}\kappa)+\omega_E(\kappa/3)}.$$

□

Disclosure probability given order flow. Because $D=1$ if and only if $q=+1$, Bayes' rule implies for any $X \in \{-2,-1,0,1,2\}$,

$$\mathbb{P}(D=1 \mid X) = \frac{\omega_P\,\mathbb{P}(X \mid q=+1)}{\omega_E\,\mathbb{P}(X \mid q=-1)+(\omega_H+\omega_Q)\,\mathbb{P}(X \mid q=0)+\omega_P\,\mathbb{P}(X \mid q=+1)}.$$

Since $X=q+z$, $\mathbb{P}(X \mid q=+1) = \mathbb{P}(z=X-1)$, $\mathbb{P}(X \mid q=0) = \mathbb{P}(z=X)$, and $\mathbb{P}(X \mid q=-1) = \mathbb{P}(z=X+1)$, hence

$$\mathbb{P}(D=1 \mid X) = \frac{\omega_P\,\mathbb{P}(z=X-1)}{\omega_E\,\mathbb{P}(z=X+1)+(\omega_H+\omega_Q)\,\mathbb{P}(z=X)+\omega_P\,\mathbb{P}(z=X-1)}.$$

In particular, bounded support implies $\mathbb{P}(D=1 \mid X=2)=1$ and $\mathbb{P}(D=1 \mid X \in \{-2,-1\})=0$, while $X \in \{0,1\}$ mix across disclosure states.

B.4 Disclosed-Branch Invariance

Proof. Consider any equilibrium under the Standing Assumptions and suppose the disclosed branch is reached (i.e., $\mathbb{P}(D=1)>0$). By the feasible action set and the disclosure rule, $D=1$ occurs if and only if the blockholder chooses Public Voice $P \equiv (+1,1)$, hence $a=1$ almost surely on $\{D=1\}$ and $X=1+z$ with $z \in \{-1,0,1\}$.

Since z is independent of (v,s,ξ), conditioning on $(X,D)=(x,1)$ reveals only the noise realization $z=x-1$ and does not change beliefs about v beyond what is already implied by $D=1$. Formally, for any Borel set $B \subset \mathbb{R}$ and any $x \in \{0,1,2\}$,

$$\mathbb{P}(v \in B \mid X=x, D=1) = \mathbb{P}(v \in B \mid D=1).$$

In particular, $\pi(x,1) = \mathbb{P}(a=1 \mid X=x, D=1) = 1$ and $\mathbb{E}[v \mid X=x, D=1] = \mathbb{E}[v \mid D=1] = \mu_P$ for all $x \in \{0,1,2\}$.

Therefore, for each $x \in \{0,1,2\}$ the feed-forward pricing equation (5) on the disclosed branch evaluates identically: it depends on (X,D) only through $\hat{V}(X,D)$, $\bar{m}(X,D)$, and $p(X,1)$, all of which are strictly constant across x when $D=1$. Thus, $P_{\text{post}}(x,1)$ and the conditional bid probability are exactly the same for all $x \in \{0,1,2\}$ on the disclosed branch. □

B.5 Derivation of Conditional Means

Lemma 3 (Truncated normal expectations)**.** *Let $S \sim \mathcal{N}(\mu, \sigma^2)$ and define standardized thresholds $\alpha = (a-\mu)/\sigma$ and $\gamma = (b-\mu)/\sigma$. Then*

$$\mathbb{E}[S \mid S < a] = \mu - \sigma \frac{\phi(\alpha)}{\Phi(\alpha)},$$
$$\mathbb{E}[S \mid a \leq S < b] = \mu + \sigma \frac{\phi(\alpha) - \phi(\gamma)}{\Phi(\gamma) - \Phi(\alpha)},$$
$$\mathbb{E}[S \mid S \geq b] = \mu + \sigma \frac{\phi(\gamma)}{1 - \Phi(\gamma)}.$$

Proof. These are standard formulas for the mean of a truncated normal distribution. For completeness, let $Z \sim \mathcal{N}(0,1)$ and note that $\mathbb{E}[S \mid \cdot] = \mu + \sigma\,\mathbb{E}[Z \mid \cdot]$. For the left tail,

$$\mathbb{E}[Z \mid Z < \alpha] = \frac{\int_{-\infty}^{\alpha} z\phi(z)\,dz}{\Phi(\alpha)} = \frac{-\phi(\alpha)}{\Phi(\alpha)},$$

where the last equality follows from $\frac{d}{dz}\phi(z) = -z\phi(z)$. The right tail follows similarly. For an interval $[\alpha, \gamma)$,

$$\mathbb{E}[Z \mid \alpha \leq Z < \gamma] = \frac{\int_{\alpha}^{\gamma} z\phi(z)\,dz}{\Phi(\gamma) - \Phi(\alpha)} = \frac{\phi(\alpha) - \phi(\gamma)}{\Phi(\gamma) - \Phi(\alpha)}.$$

□

Derivation of $\mu_E, \mu_H, \mu_Q, \mu_P$. By iterated expectations,

$$\mathbb{E}[v \mid s \in \mathcal{S}] = \mathbb{E}[\mathbb{E}[v \mid s] \mid s \in \mathcal{S}] = \mathbb{E}[\hat{v}(s) \mid s \in \mathcal{S}],$$

for any measurable set $\mathcal{S}$. Since $\hat{v}(s) = \mu + \beta(s - \mu)$,

$$\mathbb{E}[v \mid s \in \mathcal{S}] = \mu + \beta\big(\mathbb{E}[s \mid s \in \mathcal{S}] - \mu\big).$$

Applying Lemma 3 to $s \sim \mathcal{N}(\mu, \sigma_s^2)$ with the relevant truncation regions $\mathcal{S}$ yields the expressions for $\mu_E, \mu_H, \mu_Q, \mu_P$ in the main text. In particular:

- $\mu_E = \mathbb{E}[v \mid s < k_1]$ uses the left-tail formula with $\alpha = (k_1 - \mu)/\sigma_s$.
- $\mu_H = \mathbb{E}[v \mid k_1 \leq s < k_0]$ uses the interval formula with $\alpha = (k_1 - \mu)/\sigma_s$ and $\gamma = (k_0 - \mu)/\sigma_s$.
- $\mu_Q = \mathbb{E}[v \mid k_0 \leq s < k_D]$ uses the interval formula with $\alpha = (k_0 - \mu)/\sigma_s$ and $\gamma = (k_D - \mu)/\sigma_s$.
- $\mu_P = \mathbb{E}[v \mid s \geq k_D]$ uses the right-tail formula with $\alpha = (k_D - \mu)/\sigma_s$.

□

Derivation of $\mathbb{E}[v \mid X, D]$. For $D = 1$, only Public Voice is feasible, so $\mathbb{E}[v \mid X, 1] = \mu_P$ for all compatible $X \in \{0, 1, 2\}$.

For $D = 0$, Exit, Hold, and Quiet Voice are feasible. Since $X = q + z$ with z independent of (v, s), the conditional distribution of v given (X, D) is obtained by mixing the action-specific conditional means using Bayes weights proportional to the ex ante action probabilities and the likelihoods $\mathbb{P}(X \mid q)$. Note that Hold (H) and Quiet Voice (Q) both have $q = 0$ and hence generate identical order-flow distributions. Using Bayes' rule and the conditional means (μ_E, μ_H, μ_Q) gives:

- $X = -2$ implies Exit, hence $\mathbb{E}[v \mid -2, 0] = \mu_E$.
- $X = 1$ implies $q = 0$ and $z = +1$, so

$$\mathbb{E}[v \mid 1, 0] = \frac{\omega_H \mu_H + \omega_Q \mu_Q}{\omega_H + \omega_Q}.$$

- $X = -1$ mixes Exit ($z = 0$), Hold ($z = -1$), and Quiet Voice ($z = -1$). Writing the Bayes weights as

$$w_E = \frac{\omega_E\, p_0}{\mathcal{D}_{-1}}, \quad w_H = \frac{\omega_H\, p_1}{\mathcal{D}_{-1}}, \quad w_Q = \frac{\omega_Q\, p_1}{\mathcal{D}_{-1}},$$

  with $\mathcal{D}_{-1} = (\omega_H + \omega_Q)p_1 + \omega_E\, p_0$, yields $\mathbb{E}[v \mid -1, 0] = w_E \mu_E + w_H \mu_H + w_Q \mu_Q$.
- $X = 0$ mixes Quiet Voice/Hold with $z = 0$ and Exit with $z = 1$. Writing the Bayes weights as

$$w_E = \frac{\omega_E\, p_1}{\mathcal{D}_0}, \quad w_H = \frac{\omega_H\, p_0}{\mathcal{D}_0}, \quad w_Q = \frac{\omega_Q\, p_0}{\mathcal{D}_0},$$

  with $\mathcal{D}_0 = (\omega_H + \omega_Q)p_0 + \omega_E\, p_1$, yields $\mathbb{E}[v \mid 0, 0] = w_E \mu_E + w_H \mu_H + w_Q \mu_Q$.

□

B.6 Proof of Proposition 3

Proof. The secondary market sets $P_{\text{post}}(X, D) = \delta\, \mathbb{E}[Y \mid X, D]$. Conditional on the fully revealed regulatory state (X, D), the expected standalone value $\hat{V}(X, D)$ and expected premium wedge $\bar{m}(X, D)$ are constants. The bid indicator $\mathbf{1}\{\text{bid}\}$ is conditionally independent of (v, a) given (X, D), so $\mathbb{E}[\mathbf{1}\{\text{bid}\} \mid X, D] = p(X, D)$.

In the takeover payoff, the expected required payout is $\hat{V}(X, D) + \bar{m}(X, D)$. Using conditional independence,

$$\mathbb{E}[(1 - \mathbf{1}\{\text{bid}\}) \cdot (v + a\tilde{\Delta}) \mid X, D] = (1 - p(X, D))\big(\mathbb{E}[v \mid X, D] + \tilde{\Delta}\pi(X, D)\big) = (1 - p(X, D))\hat{V}(X, D).$$

Taking expectations of Y conditional on (X, D) yields

$$\mathbb{E}[Y \mid X, D] = p(X, D) \cdot (\hat{V}(X, D) + \bar{m}(X, D)) + (1 - p(X, D)) \cdot \hat{V}(X, D).$$

This algebraically simplifies to $\hat{V}(X, D) + p(X, D)\bar{m}(X, D)$. Multiplying by δ gives the exact decomposition for $P_{\text{post}}(X, D)$.

For the pre-disclosure anonymous execution price, the market maker prices the firm using the law of iterated expectations: $\mathbb{E}[Y \mid X] = \mathbb{E}[\mathbb{E}[Y \mid X, D] \mid X]$. Because $P_{\text{post}}(X, D) = \delta \mathbb{E}[Y \mid X, D]$, substituting the conditional expectation yields $P_{\text{trade}}(X) = \sum_d \mathbb{P}(D = d \mid X) P_{\text{post}}(X, d)$, completing the proof. □

B.7 Proof of Bid Probability Monotonicity

Proof. Let

$$T(X,D) \equiv \frac{\hat{V}(X,D) + \bar{m}(X,D) + K - (\bar{S} + \pi(X,D)\Delta_S)}{s_\xi}.$$

Then the unconditional bid probability is $p(X,D) = \lambda_B(1 - \Lambda(T(X,D)))$.

Differentiate T with respect to π. Since

$$\partial_\pi \hat{V}(X,D) = \tilde{\Delta}, \qquad \partial_\pi \bar{m}(X,D) = \tilde{m} - m_0.$$

Therefore:

$$\frac{\partial T(X,D)}{\partial \pi(X,D)} = \frac{\tilde{\Delta} + (\tilde{m} - m_0) - \Delta_S}{s_\xi}.$$

Under Assumption (A5), $\tilde{\Delta} + \tilde{m} - m_0 > \Delta_S$, making this derivative strictly positive. Since $\Lambda' = \lambda > 0$, we have:

$$\frac{\partial p(X,D)}{\partial \pi(X,D)} = -\lambda_B \lambda(T(X,D)) \cdot \frac{\tilde{\Delta} + \tilde{m} - m_0 - \Delta_S}{s_\xi} < 0.$$

Thus, higher inferred engagement strictly deters bids through the efficient pricing of the standalone target fundamental value. □

B.8 Proof of Within-Regime Liquidity Effects

Proof. Fix $(\omega_E, \omega_H, \omega_Q)$ and write $p_0 \equiv 1 - \frac{2}{3}\kappa$ and $p_1 \equiv \frac{\kappa}{3}$. For any reached nondisclosed information set (so denominators are nonzero), Proposition 2 gives:

$$\pi(1,0) = \frac{\omega_Q}{\omega_H + \omega_Q}, \qquad \pi(-1,0) = \frac{\omega_Q p_1}{(\omega_H + \omega_Q)p_1 + \omega_E p_0}, \qquad \pi(0,0) = \frac{\omega_Q p_0}{(\omega_H + \omega_Q)p_0 + \omega_E p_1}.$$

The first expression does not involve κ, hence $\partial \pi(1,0)/\partial \kappa = 0$.

For $X = -1$, differentiating with respect to κ yields

$$\frac{\partial \pi(-1,0)}{\partial \kappa} = \frac{(\omega_Q/3)\omega_E}{\left((\omega_H + \omega_Q)(\kappa/3) + \omega_E(1 - \frac{2}{3}\kappa)\right)^2} \geq 0,$$

with strict inequality whenever $\omega_E \omega_Q > 0$.

For $X = 0$, differentiating yields

$$\frac{\partial \pi(0,0)}{\partial \kappa} = -\frac{(\omega_Q/3)\omega_E}{\left((\omega_H + \omega_Q)(1 - \frac{2}{3}\kappa) + \omega_E(\kappa/3)\right)^2} \leq 0,$$

again strict whenever $\omega_E \omega_Q > 0$.

Finally, $\pi(-2,0) = 0$ and $\pi(X,1) = 1$ follow from Proposition 2. The claims for $\bar{m}(X,D) = m_0 + (\tilde{m} - m_0)\pi(X,D)$ follow because m is affine in π with slope $\tilde{m} - m_0 > 0$ under Assumption (A3). □

B.9 Proof of Takeover Comparative Statics

Proof. Recall the unconditional bid probability function

$$p(X,D)=\lambda_B\left(1-\Lambda\left(\frac{\hat{V}(X,D)+\bar{m}(X,D)+K-(\bar{S}+\pi(X,D)\Delta_S)}{s_\xi}\right)\right).$$

Holding (X,D) fixed, $\partial p/\partial\bar{S}>0$ and $\partial p/\partial K<0$ follow immediately from the properties of the logistic CDF $\Lambda'=\lambda>0$. Because the bid probability is anchored to $\hat{V}(X,D)$ rather than the anticipatory market prices $P_{\text{trade}}(X)$ or $P_{\text{post}}(X,D)$, the price-to-entry feedback loop is eliminated entirely, rendering arbitrary price regularity conditions unnecessary. □

B.10 Proof of Equilibrium Cutoff Equations

Proof. Define the action-specific expected payoffs by evaluating $U(q,a\mid s)$ against the anonymous execution price $P_{\text{trade}}(X)$ for the four actions in Proposition 1:

$$\begin{aligned}U_E&\equiv U(-1,0\mid s)=\mathbb{E}_z[P_{\text{trade}}(-1+z)],\\U_H(s)&\equiv U(0,0\mid s),\\U_Q(s)&\equiv U(0,1\mid s),\\U_P(s)&\equiv U(+1,1\mid s).\end{aligned}$$

Because $U_E=\mathbb{E}_z[P_{\text{trade}}(-1+z)]$, the expected payoff to Exit is perfectly invariant to the blockholder's private signal s. In a cutoff strategy equilibrium with thresholds $k_1<k_0<k_D$, the blockholder is indifferent between adjacent actions at the boundary signals where the action switches. The three indifference conditions are:

- At $s=k_1$: Exit vs. Hold, i.e., $U_E=U_H(k_1)$.
- At $s=k_0$: Hold vs. Quiet Voice, i.e., $U_H(k_0)=U_Q(k_0)$.
- At $s=k_D$: Quiet Voice vs. Public Voice, i.e., $U_Q(k_D)=U_P(k_D)$.

These are exactly the conditions (7)–(9). □

B.11 Proof of Minority Takeover Gains Decomposition

Proof. Minority takeover gains are defined as $\Delta^{\min}(\kappa)=\mathbb{E}\big[m^R(a)\cdot\mathbf{1}\{\text{bid}\}\big]$, where $m^R(a)=m_0+a(\tilde{m}-m_0)$. Conditioning on (X,D) and using the fact that, conditional on public information (X,D), the bid indicator (which depends on the bidder's independent synergy shock ξ) is independent of the realized engagement a:

$$\mathbb{E}\big[m^R(a)\cdot\mathbf{1}\{\text{bid}\}\mid X,D\big]=\mathbb{E}[m^R(a)\mid X,D]\cdot\mathbb{E}[\mathbf{1}\{\text{bid}\}\mid X,D]=\bar{m}(X,D)\cdot p(X,D).$$

Taking expectations over (X,D) yields

$$\Delta^{\min}(\kappa)=\mathbb{E}\big[\bar{m}(X,D)\cdot\mathbf{1}\{\text{bid}\}\big].$$

Using $\bar{m}(X,D) = m_0 + (\tilde{m} - m_0)\pi(X,D)$ and linearity of expectation,

$$\begin{aligned}\Delta^{\min}(\kappa) &= \mathbb{E}\big[(m_0 + (\tilde{m} - m_0)\pi(X,D)) \cdot \mathbf{1}\{\text{bid}\}\big] \\ &= m_0 \cdot \mathbb{E}[\mathbf{1}\{\text{bid}\}] + (\tilde{m} - m_0) \cdot \mathbb{E}\big[\pi(X,D) \cdot \mathbf{1}\{\text{bid}\}\big],\end{aligned}$$

which is the claimed decomposition with $\mathbb{P}(\text{bid}) = \mathbb{E}[\mathbf{1}\{\text{bid}\}]$. □

B.12 Proof of Proposition 4 (Existence under Maintained Regularity Conditions)

Proof. Let Θ be the compact ordered cutoff domain from Proposition 4. By assumption, the cutoff mapping $T : (k_1, k_0, k_D) \mapsto (k_1', k_0', k_D')$ is well defined on Θ, continuous on reached information sets, and maps Θ into itself.

The mapping T is generated by explicitly computing feed-forward prices via (5) and (6). Cutoffs map continuously to action probabilities ω_a. For any reached information set, the relevant Bayes denominator is strictly positive by definition of "reached," so the posteriors $\pi(X,D)$ and conditional disclosure probabilities $\mathbb{P}(D = d|X)$ are well defined. Off path, beliefs can be completed arbitrarily without affecting the equilibrium allocation. These beliefs map into P_{post} and, via linear combination, into P_{trade}. Since Θ is a compact, convex subset of a Euclidean space, Brouwer's Fixed-Point Theorem guarantees that T has at least one fixed point in Θ. □

B.13 Numerical Regularity and Fixed-Point Selection

Assumption (A6) is used as a numerical regularity condition on the cutoff mapping T. Analytically bounding the spectral radius of the Jacobian of T is intractable because the derivatives involve highly nonlinear interactions between inverse Mills ratios, the logistic CDF of the bidder's entry rule, and Bayesian posterior updating. Following the methodological precedent established for discrete order-flow feedback models (Edmans, Goldstein, et al., 2015), I therefore rely on numerical verification rather than a theorem: in all numerical exercises presented in this paper, iterating T from multiple starting points converges to the same fixed point. This is evidence of stable fixed-point selection, not a proof of global uniqueness.

B.14 Proof of Lemma 2 (Endpoint Behavior)

Proof. Fix a cutoff strategy and hence $(\omega_E, \omega_H, \omega_Q, \omega_P)$.

(i) Left endpoint ($\kappa \downarrow 0$). As $\kappa \downarrow 0$, we have $\mathbb{P}(z = 0) = 1 - \frac{2}{3}\kappa \to 1$ and $\mathbb{P}(z = \pm 1) = \frac{\kappa}{3} \to 0$. Therefore $z = 0$ almost surely in the limit and $X = q + z \to q$. Since $D = 1$ if and only if $q = +1$, the disclosure state becomes a deterministic function of X on all reached order-flow states: $X = 1$ implies $(q, D) = (+1, 1)$, while $X \in \{-1, 0\}$ implies $D = 0$. Hence the conditional mixing weights in (2) satisfy $\mathbb{P}(D = d \mid X) \to \mathbf{1}\{d = D(X)\}$, and

$$P_{\text{trade}}(X) \;\longrightarrow\; P_{\text{post}}(X, D(X)).$$

(ii) Right endpoint ($\kappa \uparrow 1$). As $\kappa \to 1$, $\mathbb{P}(z = -1) = \mathbb{P}(z = 0) = \mathbb{P}(z = 1) = 1/3$. Because z has bounded support, the extreme order flows identify q and hence D: if $X = 2$, then necessarily

$(q, z) = (+1, +1)$ and thus $D = 1$, so $\mathbb{P}(D = 1 \mid X = 2) = 1$; if $X = -2$, then necessarily $(q, z) = (-1, -1)$ and thus $D = 0$; if $X = -1$, then $q = +1$ would require $z = -2$ which is impossible, so $\mathbb{P}(D = 1 \mid X = -1) = 0$. For $X \in \{0, 1\}$ both $q = 0$ and $q = +1$ are feasible with appropriate noise realizations, so $\mathbb{P}(D = 1 \mid X) \in (0, 1)$ is determined by Bayes' rule. A fully uninformative limit for all (X, D) does not obtain under bounded noise. $\square$

B.15 Proof of Proposition 5 (Nonmonotonicity)

Proof. The analytic decomposition $\Delta^{\min}(\kappa) = \Delta^{\text{base}}(\kappa) + \Delta^{\text{act}}(\kappa)$ is established in Appendix B.11. The monotonicity of $\Delta^{\text{base}}(\kappa)$ and $\Delta^{\text{act}}(\kappa)$ in κ is a general-equilibrium comparative static that depends on the endogenous cutoff vector (k_1, k_0, k_D) and the induced belief and pricing system. We therefore establish the hump shape and the monotonic component patterns numerically: solving the equilibrium cutoff system over the liquidity grid in Section 5 yields an interior maximizer of $\Delta^{\min}(\kappa)$ in the baseline calibration. The sensitivity exercises in Section 5 illustrate how the peak location and curvature can shift as primitives vary. $\square$

B.16 Derivations for Section 8

(ND) No disclosure. Under the no-disclosure regime $S_{\text{ND}} = \emptyset$, the market conditions only on $X = q + z$ with $\mathbb{P}(z = 0) = p_0 = 1 - \frac{2}{3}\kappa$ and $\mathbb{P}(z = \pm 1) = p_1 = \frac{\kappa}{3}$. Writing $\mathbb{P}(X \mid q)$ for the order-flow likelihood induced by z, Bayes' rule implies

$$\pi_{\text{ND}}(X) \equiv \mathbb{P}(a = 1 \mid X) = \frac{\omega_Q \mathbb{P}(X \mid q = 0) + \omega_P \mathbb{P}(X \mid q = +1)}{\omega_E \mathbb{P}(X \mid q = -1) + (\omega_H + \omega_Q)\mathbb{P}(X \mid q = 0) + \omega_P \mathbb{P}(X \mid q = +1)}.$$

Substituting $\mathbb{P}(X = q) = p_0$ and $\mathbb{P}(X = q \pm 1) = p_1$ (and zero otherwise) yields the closed-form expressions reported in Section 8.

(NR) Baseline stake disclosure plus a noisy rumor. Under $S_{\text{NR}} = (D, R)$ with $D = \mathbf{1}\{q = +1\}$, the disclosed branch $D = 1$ pins down Public Voice in the baseline action set, so $\pi(X, 1, R) = 1$ for all feasible X. When $D = 0$, $q \in \{-1, 0\}$ and the rumor has hit/false-alarm rates $\mathbb{P}(R = 1 \mid q = 0, a = 1) = \eta_1$ and $\mathbb{P}(R = 1 \mid q = 0, a = 0) = \eta_0$, with the convention $\mathbb{P}(R = 1 \mid q = -1) = \eta_0$. Bayes' rule gives, for each feasible (X, R),

$$\begin{aligned}
\pi(X, 0, R) &= \mathbb{P}(Q \mid X, 0, R) \\
&= \frac{\omega_Q \, \mathbb{P}(X \mid q = 0)\mathbb{P}(R \mid Q)}{\omega_E \, \mathbb{P}(X \mid q = -1)\mathbb{P}(R \mid E) + \mathbb{P}(X \mid q = 0)\big(\omega_H \, \mathbb{P}(R \mid H) + \omega_Q \, \mathbb{P}(R \mid Q)\big)}.
\end{aligned}$$

Evaluating $\mathbb{P}(X \mid q)$ using $X = q + z$ yields the cases in Section 8: $X = -2$ is degenerate (Exit only); $X = 1$ pins down $q = 0$ so κ drops out; and $X \in \{-1, 0\}$ mixes $q = -1$ and $q = 0$ with weights proportional to (p_0, p_1), generating the displayed formulas.

C Tables

D	Order flow X	$P(X,D)$	$\pi(X,D)$	$p(X,D)$	$\bar{m}(X,D)$
0	-2	0.51	0.00	0.18	0.10
0	-1	0.68	0.07	0.15	0.12
0	0	0.92	0.18	0.07	0.16
0	1	0.96	0.20	0.05	0.16
1	0	1.55	1.00	0.00	0.42
1	1	1.55	1.00	0.00	0.42
1	2	1.55	1.00	0.00	0.42

Table C.1: Baseline numerical equilibrium (values rounded to two decimals). Parameters: $\mu = 1$, $\sigma_v = 0.5$, $\sigma_\varepsilon = 0.5$, $\delta = 0.95$, $\Delta = 0.25$, $C_0 = 0.25$, $\chi = 0.5$, $\kappa = 0.5$, $m_0 = 0.10$, $m_1 = 0.45$, $\bar{S} = 1.10$, $\Delta_S = 0.30$, $K = 0.15$, $s_\xi = 0.15$, $\lambda_B = 0.20$, $\rho = 0.9$. Cutoffs: k_1, k_0, and k_D (see updated numerical output).

X	$\pi(X,0)$	$\pi_{\mathrm{ND}}(X)$	$\pi_{\mathrm{NR}}(X,0,R=0)$	$\pi_{\mathrm{NR}}(X,0,R=1)$
-1	0.07	0.07	0.03	0.19
0	0.18	0.28	0.07	0.40
1	0.20	0.77	0.08	0.43

(q,a)	π_{FI}
$(-1,0)$	0.00
$(0,0)$	0.00
$(0,1)$	1.00
$(1,1)$	1.00

Table C.2: Illustration of posterior engagement probabilities under alternative disclosure signals (Section 8), evaluated at the baseline calibrated equilibrium cutoffs and $\kappa = 0.5$. For NR, the rumor parameters are set to $(\eta_1, \eta_0) = (0.75, 0.25)$.

Parameter	Symbol	Value	Interpretation
Fundamentals			
Prior mean	μ	1.00	Normalized baseline value
Fundamental volatility	σ_v	0.50	Moderate uncertainty
Signal noise	σ_ε	0.50	Informative but noisy signal
Engagement			
Base engagement cost	C_0	0.25	Sufficiently high to restore passive Hold
Cost sensitivity	χ	0.50	Moderate signal-dependence
Success probability	ρ	0.90	High success rate
Value improvement	Δ	0.25	25% improvement if successful
Takeover			
Base premium	m_0	0.10	10% premium without activism
Activism premium	m_1	0.45	45% premium with activism
Baseline synergy	$\bar{S}$	1.10	Balanced to ensure highly sensitive bid variation
Synergy improvement	Δ_S	0.30	Activism increases synergy
Bidding cost	K	0.15	Fixed deal friction
Synergy scale (Logistic)	s_ξ	0.15	Controls probability dispersion
Bidder arrival rate	λ_B	0.20	Calibrates unconditional bid rates to 2-8%
Other			
Discount factor	δ	0.95	$\delta = \exp(-r\tau)$ between pricing and settlement
Noise intensity	κ	0.50	Baseline (varies in analysis)

Table C.3: Baseline parameterization for numerical analysis.

D Figures

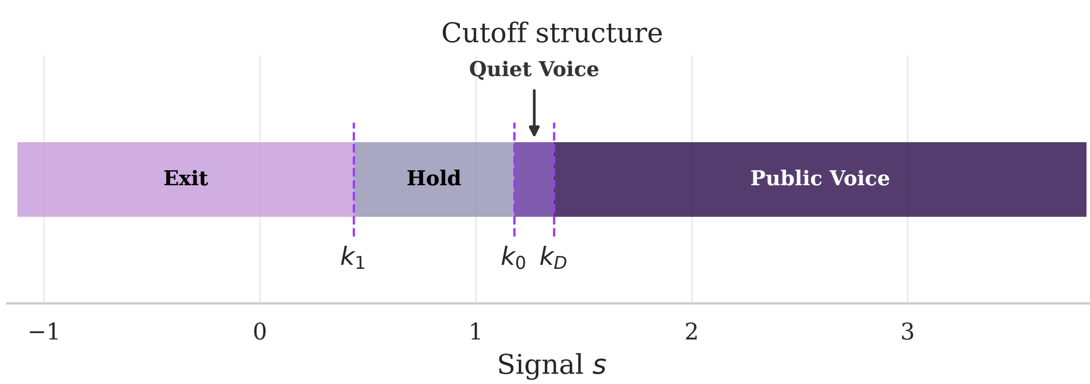

Figure D.1: Equilibrium cutoff structure. The blockholder follows a threshold strategy with weakly ordered cutoffs. In general, the signal space can be partitioned into up to four regions (Exit, Hold, Quiet Voice, Public Voice), with disclosure occurring only under Public Voice. In the baseline numerical example, all four regions are present, with a narrow Quiet Voice region (i.e., k_D lies close to k_0).

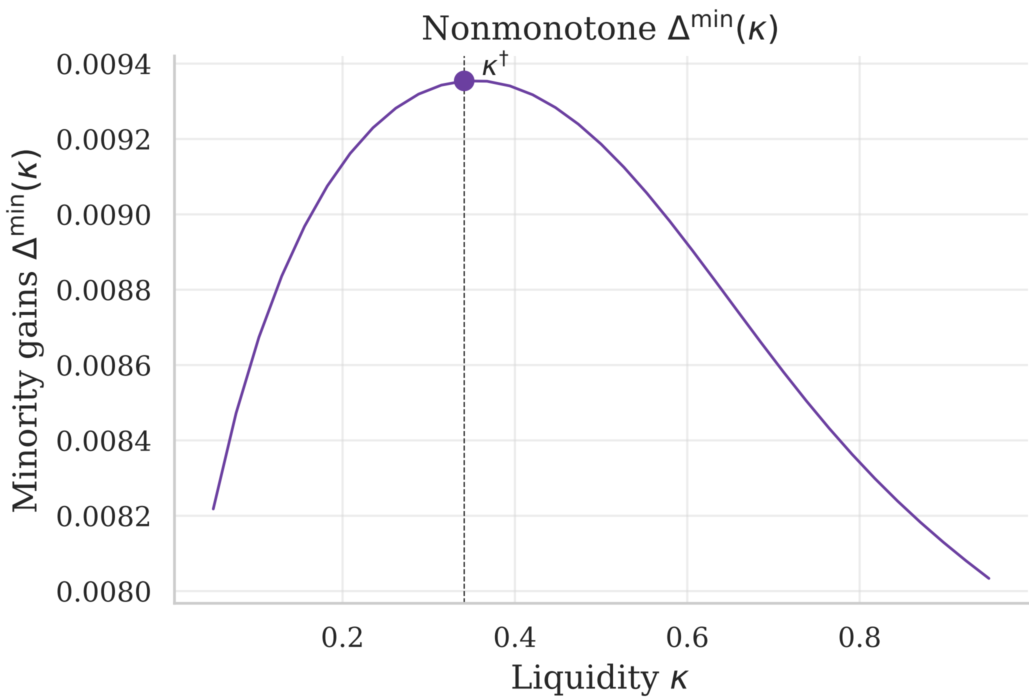

Figure D.2: Nonmonotonic effect of liquidity on expected minority takeover gains (computed over $\kappa \in [0.05, 0.95]$). The vertical dashed line marks the turning point $\kappa^{\dagger}$ (the maximizer of $\Delta^{\min}$ on the plotted grid) in the baseline calibration.

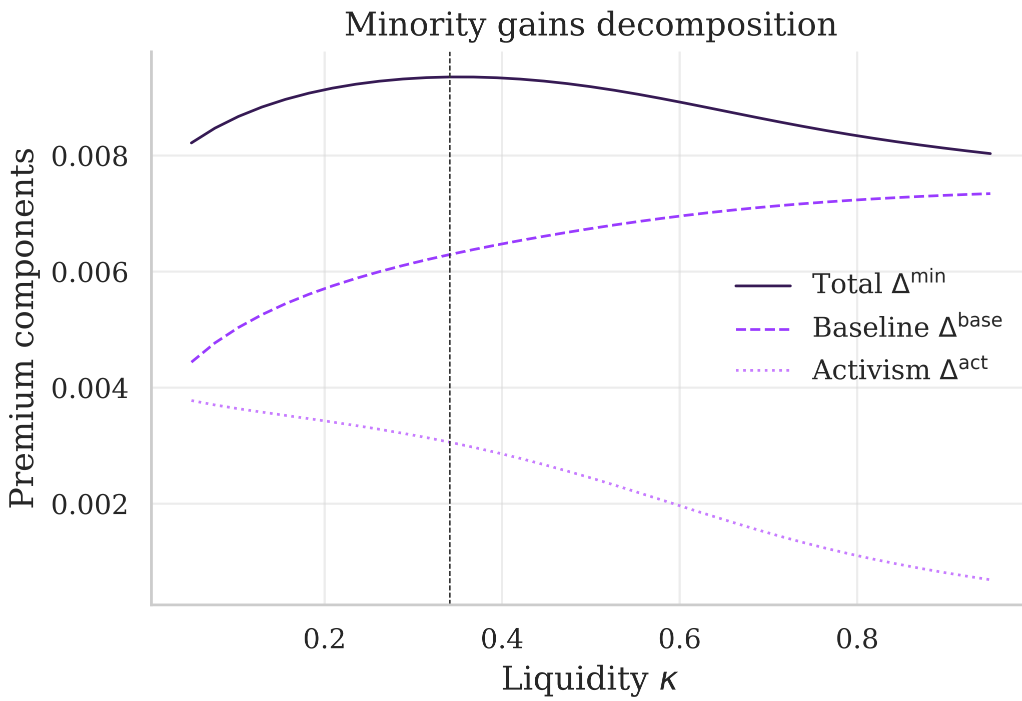

Figure D.3: Decomposition of minority takeover gains (computed over $\kappa \in [0.05, 0.95]$). The base component $m_0 \cdot \mathbb{P}(\text{bid})$ and the activism component $\Delta^{\text{act}}(\kappa)$ vary smoothly with liquidity in the baseline calibration; the dashed line marks $\kappa^{\dagger}$.

Figure D.4: Equilibrium prices by state (X, D) in the baseline numerical example. Left panel: nondisclosed states ($D = 0$). Right panel: disclosed states ($D = 1$). Bars show post-disclosure prices $P_{\text{post}}(X, D)$ and diamonds show execution prices $P_{\text{trade}}(X)$. Numbers above bars indicate the posterior engagement probability $\pi(X, D)$.

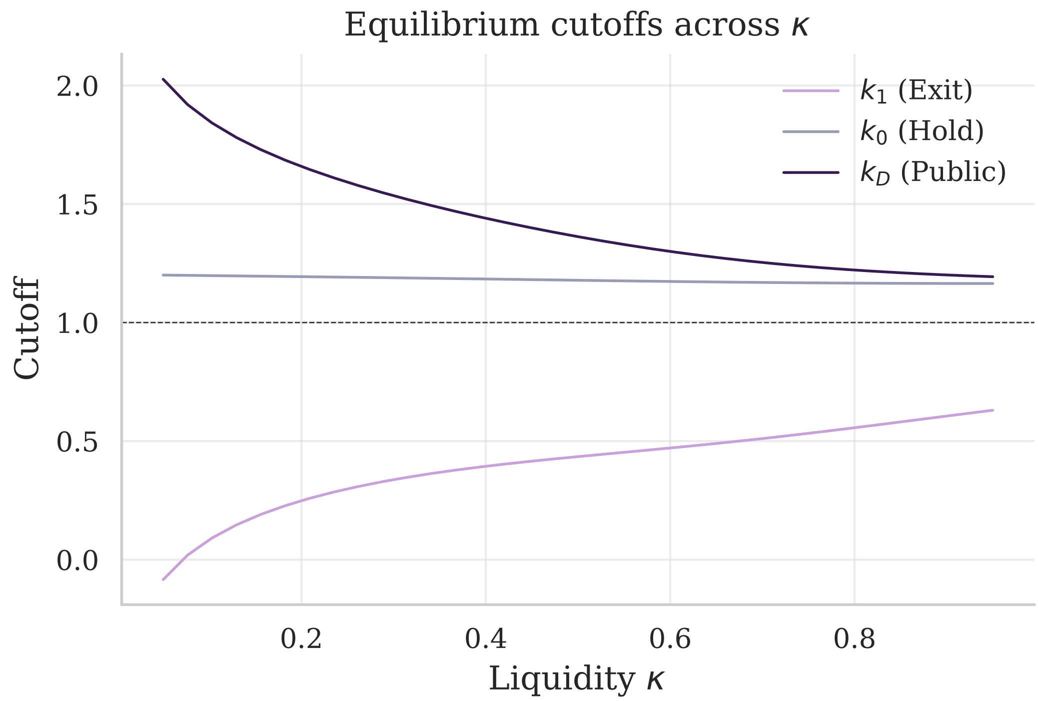

Figure D.5: Equilibrium cutoffs as functions of liquidity (computed over $\kappa \in [0.05, 0.95]$). The ordering $k_1 \leq k_0 \leq k_D$ is preserved; regions may collapse. The horizontal dashed line marks the prior mean μ.

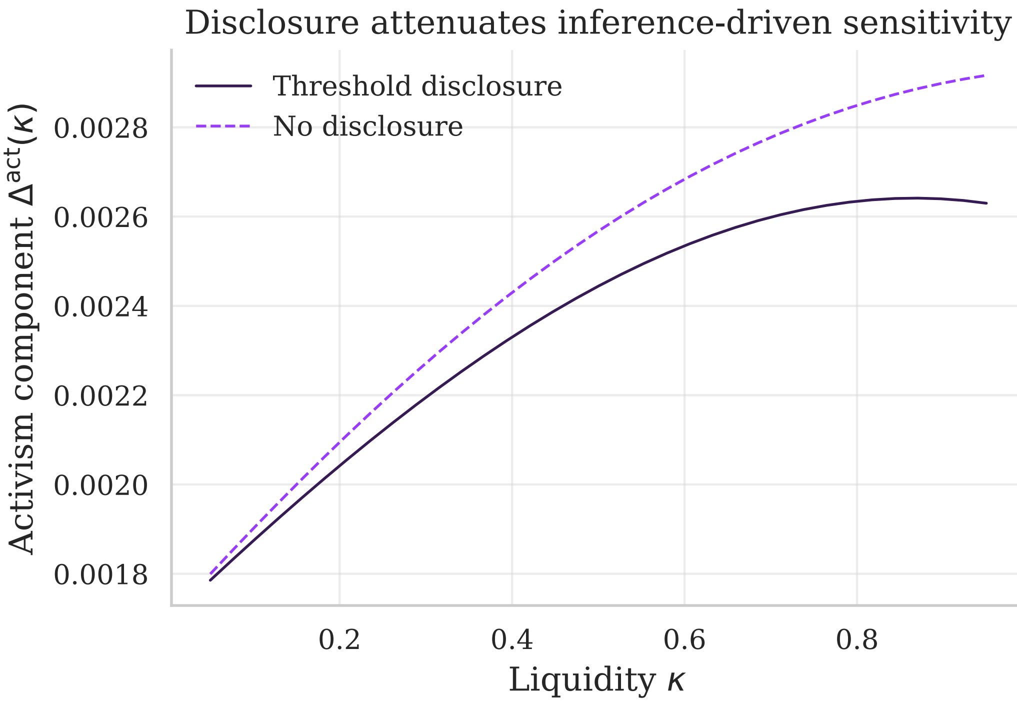

Figure D.6: Disclosure attenuation through the inference channel (partial equilibrium in κ, computed over $\kappa \in [0.05, 0.95]$). We fix the blockholder's cutoff strategy at the baseline calibrated equilibrium ($\kappa = 0.5$) and vary liquidity κ only through the order-flow noise distribution. The blue curve plots $\Delta^{\mathrm{act}}(\kappa)$ under threshold disclosure with fixed cutoffs. The orange curve plots the counterfactual "no-disclosure" benchmark with the same fixed cutoffs, in which the market conditions only on order flow X.

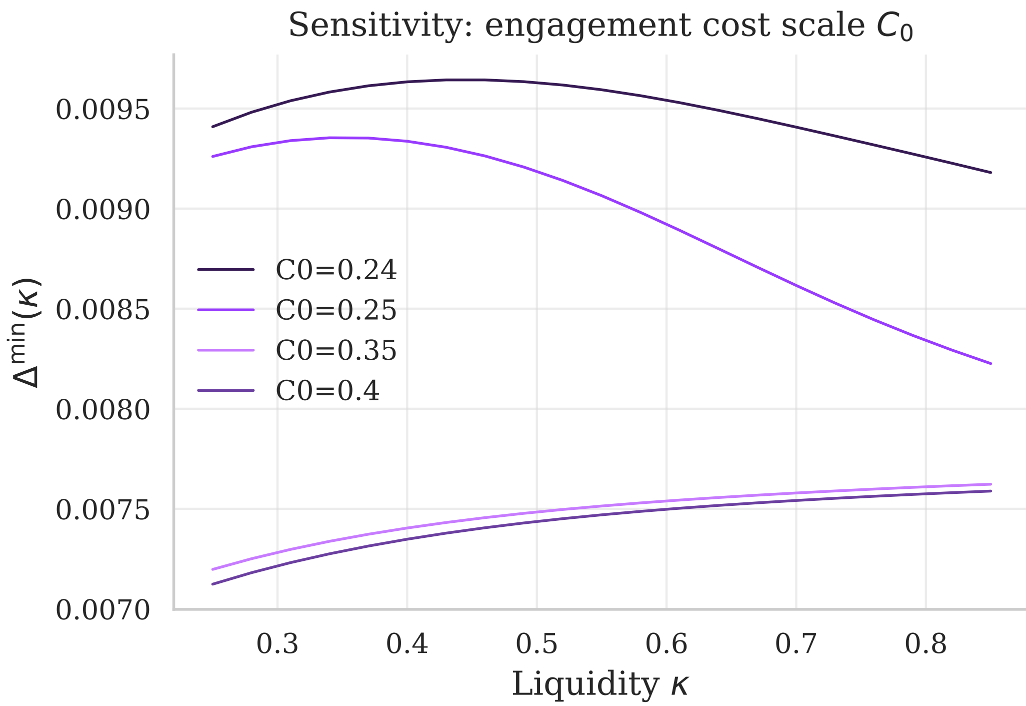

Figure D.7: Sensitivity to engagement cost C_0 (computed over $\kappa \in [0.25, 0.85]$). Lower engagement costs shift $\Delta^{\min}(\kappa)$ upward and can make it increasing on the plotted range; higher costs shrink the voice regions and can generate an interior peak.

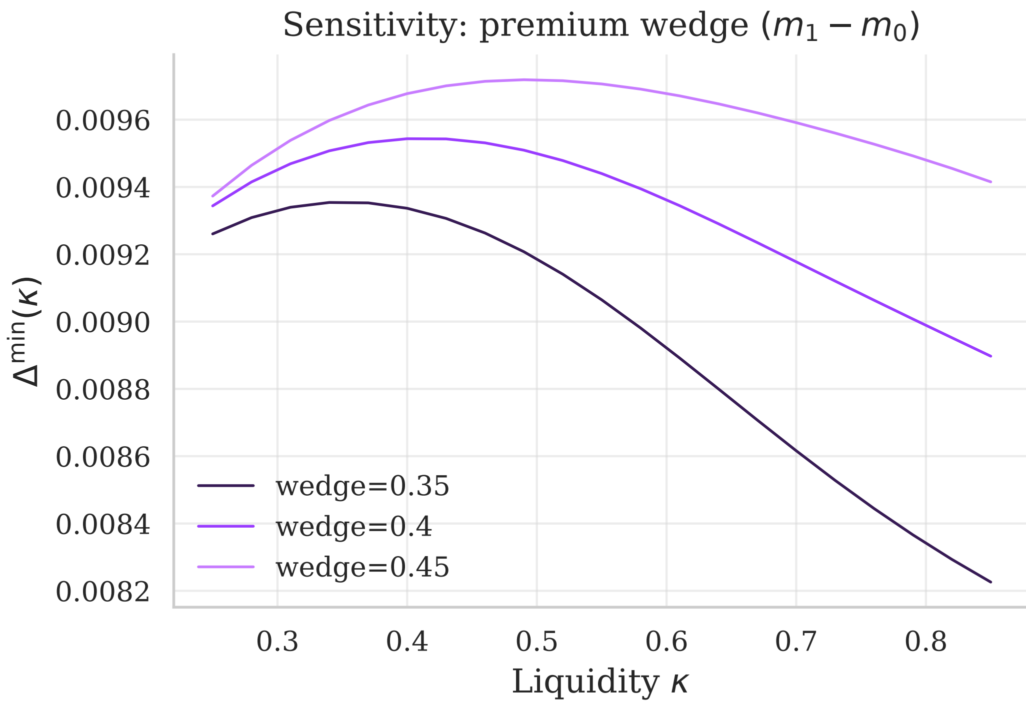

Figure D.8: Sensitivity to premium wedge ($m_1 - m_0$). A larger wedge raises the premium conditional on bidding but can deter bids; expected minority takeover gains can therefore be nonmonotone in the wedge (computed over $\kappa \in [0.25, 0.85]$).

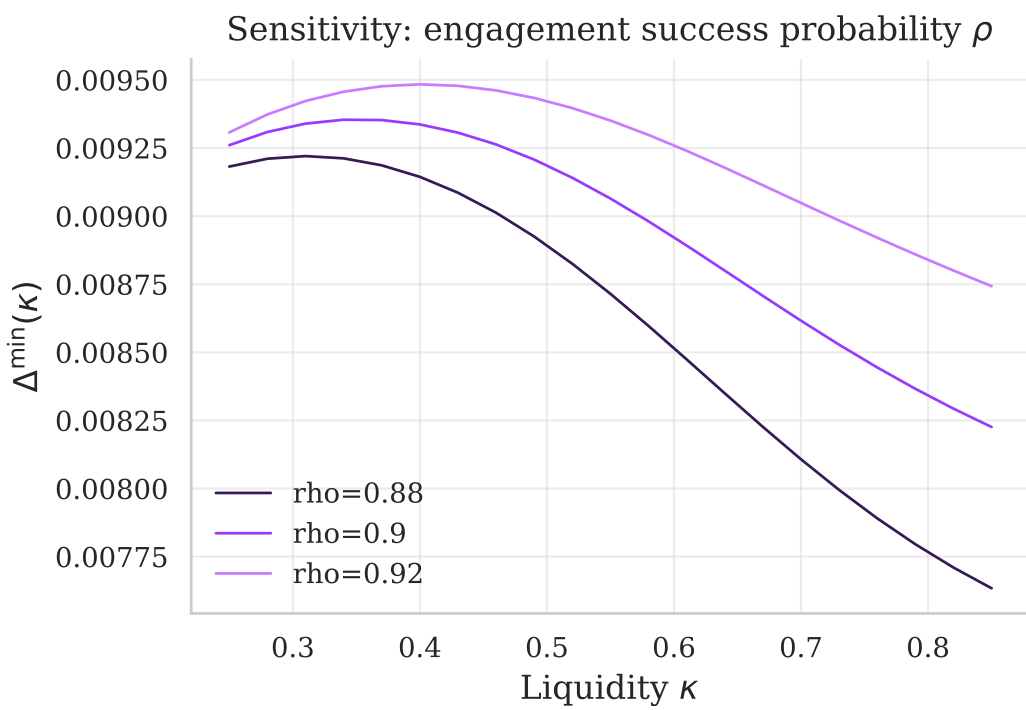

Figure D.9: Sensitivity to engagement success probability ρ (computed over $\kappa \in [0.25, 0.85]$). Lowering ρ shrinks expected engagement benefits and premia, compressing the vertical scale of $\Delta^{\min}(\kappa)$ and shifting the location of its peak on the plotted range.

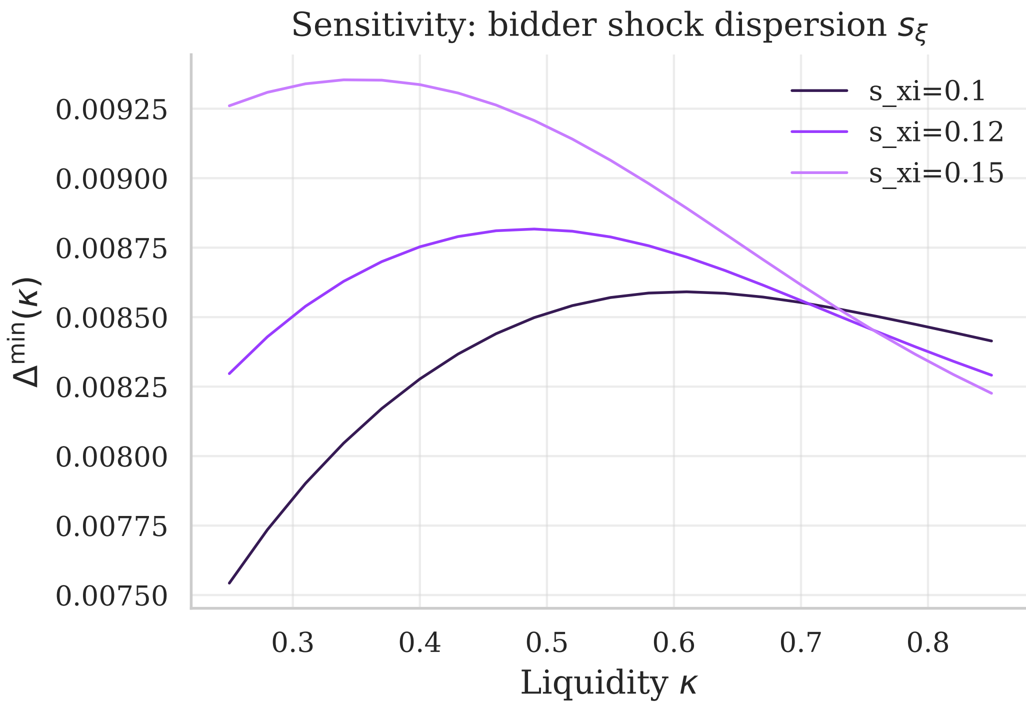

Figure D.10: Sensitivity to bidder synergy scale s_ξ (computed over $\kappa \in [0.25, 0.85]$). Changing bidder dispersion shifts bid incidence and equilibrium cutoffs, so the shape of $\Delta^{\min}(\kappa)$ can flatten or change as s_ξ varies.

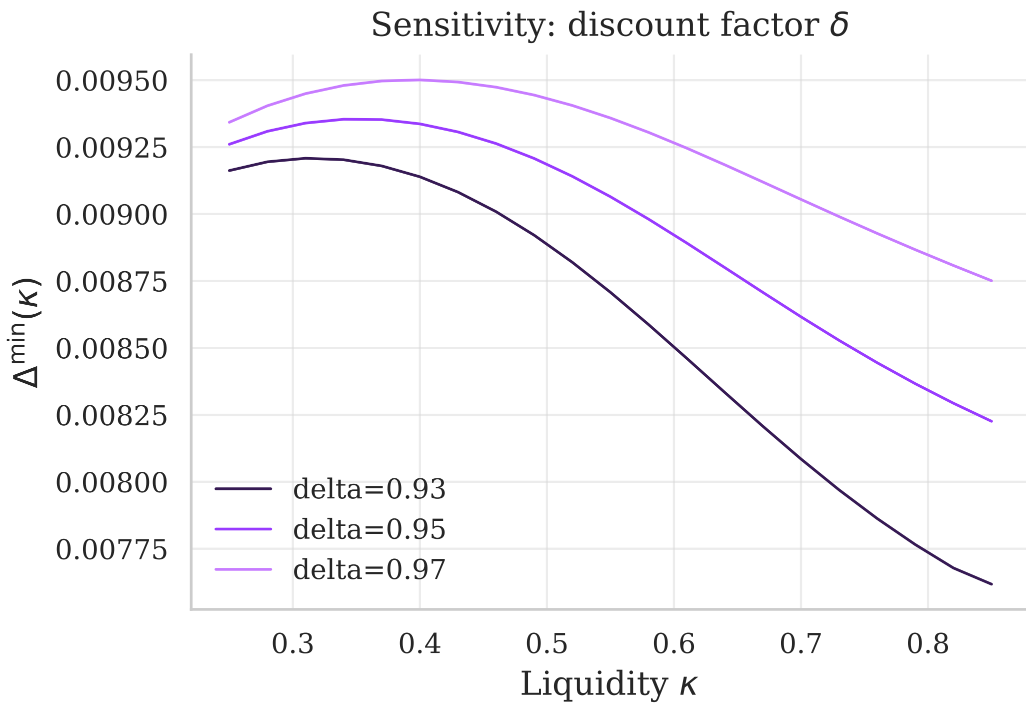

Figure D.11: Sensitivity to the discount factor δ (computed over $\kappa \in [0.25, 0.85]$). The bidder's entry probability is independent of prices, so δ affects $\Delta^{\min}(\kappa)$ only indirectly through equilibrium cutoffs and beliefs; lower δ weakens engagement incentives and can alter curvature on the plotted range.

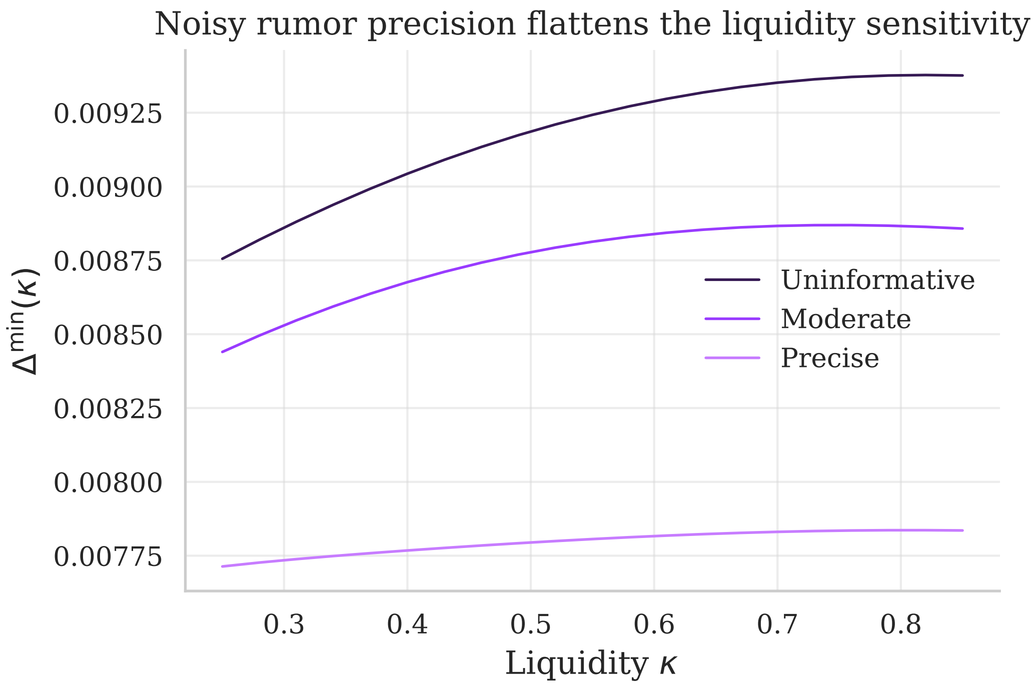

Figure D.12: Disclosure attenuation via noisy rumors (fixed cutoffs). Under the intermediate information regime, the market receives an imperfect signal of hidden engagement. As the public rumor becomes more precise ($\eta_1 - \eta_0$ increases), engagement becomes more observable in nondisclosed states, flattening the sensitivity of $\Delta^{\min}(\kappa)$ to market liquidity.

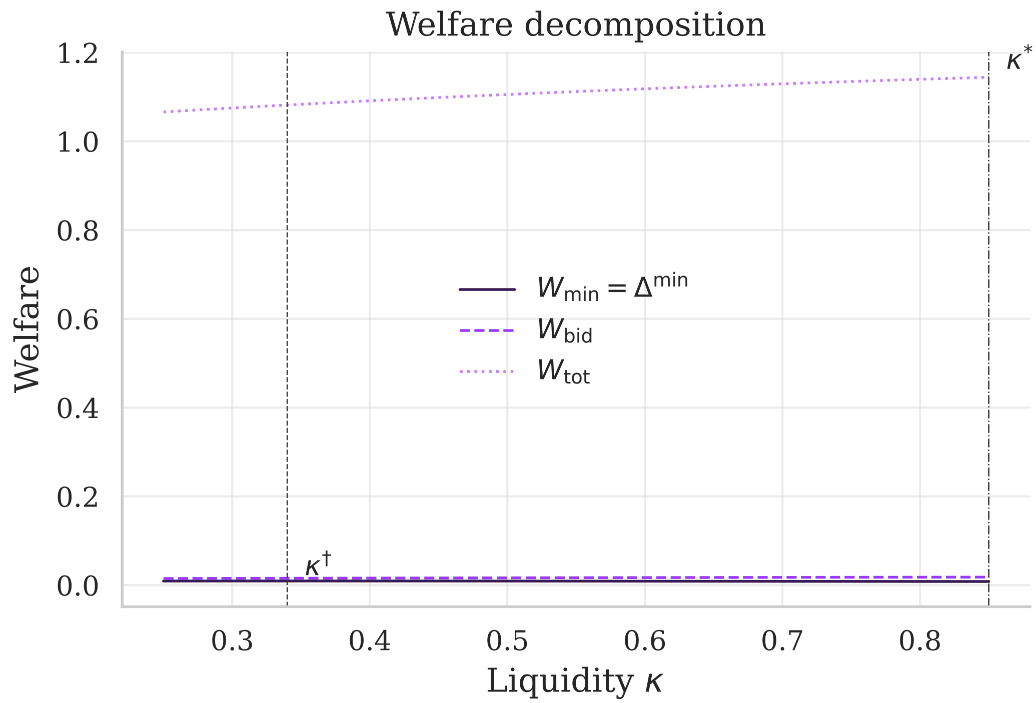

Figure D.13: Welfare decomposition and optimal liquidity. On the plotted grid, total surplus W_{tot} is increasing in κ while minority welfare $W_{\mathrm{min}} = \Delta^{\mathrm{min}}(\kappa)$ is hump-shaped. The liquidity level κ^* that maximizes total surplus can therefore differ from the level $\kappa^\dagger$ that maximizes minority takeover gains.